



Review Article

A systematic review of quantum image processing: Representation, applications and future perspectives[☆]Umar Farooq^a, Parvinder Singh^{a,*}, Atul Kumar^b^a Central University of Punjab, Bathinda, India^b Indian Institute of Technology, Jodhpur, India

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ABSTRACT

Quantum image processing uses quantum hardware to revolutionize the storage, recovery, processing, and security of quantum images across diverse applications. Although researchers have explored various facets of quantum image processing, a comprehensive systematic literature review encompassing all domains is essential for theoretical and experimental progress. This article aims to bridge this gap by systematically analyzing advancements in this field, drawing insights from a thorough review of 135 research articles published beyond 2003. Our study examines core components of quantum image processing, such as quantum image representations, advanced algorithms, transformative techniques like Quantum Fourier and Wavelet Transforms, and robust security measures. It further explores the synergy between quantum machine learning and image processing for improved classification and recognition. In addition, the study also discusses the limitations of existing research, summarizes its essential aspects, highlights gaps and challenges, and finally, provides recommendations for future research and innovations.

[☆] Abbreviations: EPR: Einstein-Podolsky-Rosen; GHZ: Greenberger-Horne-Zeilinger; FRQI: Flexible Representation of Quantum Images; NEQR: Novel Enhanced Quantum Representation; MCQI: Multi-Channel Representation for Images on Quantum Computers; QRDA: Quantum Representation of Digital Audio; FRQA: Flexible representation and manipulation of audio signals on quantum computers; QRCI: Quantum Representation model of Color digital Images; OQIM: Order-encoded Quantum Image Model; moQIM: multiple Order-encoded Quantum Image Model; MCRQI: Multi-Channel Representation for Quantum Image; QSNQ: Quantum States for N Coordinates; QSMC: Quantum States for M Colors; QUALPI: Novel Quantum representation for Log-Polar Images; CQIR: Image Representation and Processing Using Ternary Quantum Computing; FQRGI: Flexible Quantum Representation for Gray-level Images; NASS: Normal Arbitrary Superposition State; NASSRP: Normal Arbitrary Superposition State with Relative Phases; NASSTC: Normal Arbitrary Superposition State Three Components; SQR: Simple Quantum Representation of infrared images; INEQR: Improved Novel Enhanced Quantum Representation; 2DQNSA: Two-Dimensional Quantum States and Normalized Amplitude; NCQI: Novel Quantum Representation of Color Digital Images; MCQR: Red-Green-Blue Multi-Channel Quantum Representation of digital images; GNEQR: Generalized Model of NEQR; FRQCI: Flexible Representation of Quantum Color Images; BRQI: A Quantum Image Representation Based on Bitplanes; QRMMI: Quantum Representation model for Multi-Image; QRMW: Quantum Representation of Multi Wavelength images; DWT: Discrete Wavelet Transform; OCQR: Optimized Quantum Representation for Color Digital Images; QRCI: Quantum Representation Model of Color digital Images; IFRQI: Improved Flexible Representation of Quantum Images; QBIR: Quantum Block Image Representation; QIIR: Quantum Representation of Indexed Images; DQRCI: Double quantum color images representation model; FTQR: Fourier Transform Qubit Representation; INCQI: Improved Novel Quantum Image Representation; EFRQI: Enhanced FRQI; ENEQR: Enhanced NEQR; DCT-EFRQI: Direct Cosine Transform Efficient FRQI; GTQI: Geometric Transformation of Quantum Images; rGTQI: restricted Geometric Transformation of Quantum Images; CTQI: Color Transformation of Quantum Images; MC-PPT: Multi Channel - Polynomial Preparation Theorem; CFQM: Chromatic framework for Quantum Movies and applications in creating montages; MFIQC: Morphology Filtering Inspired by Quantum Collapsing; QCMF: Quantum Collapsing Median Filter; QAMF: Quantum Adaptive Median Filter; QMFSD: Quantum image Median Filtering in the Spatial Domain; QBID: Quantum Boolean Image Denoising; QBMF: Quantum Boolean Mean Filter; QCLFE: Quantum Calculus of Local Fractional Entropy; DQWT: Daubechies kernel-based Quantum Wavelet Transform; QFSD: Quantum image Filtering in the Spatial Domain; QFFD: Quantum image Filtering in the Frequency Domain; MLSONN: Multilayer Self Organizing Neural Network; QMLSONN: Quantum MLSONN; QNCDT: Wavelet denoising for quantum noise removal in chest digital tomosynthesis; XRFI: Denoising methods for X-ray fluoroscopic images; QBM: Quantum Baker Map; BQTHA: Binary image based Quantum version of the Hilditch's Algorithm; GRAPE: Gradient Ascent Pulse Engineering; PCA: Principal Component Analysis; QKD: Quantum Key Distribution; BBCE: Block Bit-plane Centrosymmetric Expansion; QFFT: quantum fractional Fourier transform; QPSO: quantum particle swarm optimization; QSVM: Quantum Support Vector Machine; QKNN: Quantum K-Nearest Neighbor; PHLA: Parametrized Hamiltonian Learning Algorithm; VQC: Variational Quantum Circuit; QCNN: Quantum Convolutional Neural Network; PQC: Parameterized Quantum Circuit; QanvNN: Qanvolutional Neural Network; QRAM: Quantum Read Access memory; DNF: Disjunctive Normal Form.

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1. Introduction

Quantum computing, an interdisciplinary field of quantum physics and information science, is a rapidly evolving research domain that is transforming the computational paradigms from classical to quantum [1,2]. Quantum computers, based on superposition, entanglement, and non-local correlations between qubits [3–5], can outperform classical computers by significantly accelerating tasks that are intractable within the classical realm [6,7]. The computational power of classical computers has historically followed Moore's Law, experiencing significant growth, but since then, due to various factors [8,9], its computing power has slowed down. In response, Feynman proposed the concept of a quantum computer to surpass classical limitations and meet the increasing computational demands of time [10]. Breakthroughs, such as Shor's algorithm for integer factorization [11], Grover's algorithm for unordered searching [12], and quantum algorithms for solving large systems of linear equations [13–15] have exemplified the power of quantum algorithms in tackling complex tasks. These advancements have directly influenced the emergence of quantum image processing with exponential speedup. Just as quantum computing has transformed cryptography and optimization, it has the potential to revolutionize image processing by enabling efficient storage, retrieval, and manipulation of high-dimensional image data through quantum algorithms [16]. The idea of quantum image processing was initiated in the early 2000s, as researchers began exploring how quantum algorithms could be adapted to process digital images. However, its foundation started with the introduction of quantum image representation models, particularly the Qubit Lattice representation in 2003 [17], which provided a structured encoding of image pixels in quantum states. Later models, such as the FRQI [18] and NEQR [19], further improved the encoding and processing of quantum images, allowing for practical quantum image transformations. Techniques like quantum DCT [8] enable image compression by converting images into frequency components to minimize the amount of information required, while QWT [20] has become a powerful tool in quantum watermarking and denoising, offering robust image compression capabilities. Further, the quantum cellular neural network [21] leverages chaotic properties, where even a few qubits can induce complex high-dimensional chaos in image pixels, providing unparalleled security. These have led to advancements in quantum image filtering, edge detection, segmentation, and encryption, tasks that are computationally demanding in classical image processing. Beyond image representation, quantum computing applications extend to cryptography [22–26], optimization [27–30], chemistry [31–33], finance [34,35], and machine learning [36–42], showcasing its transformative potential across various domains. As research in quantum image processing continues to evolve, quantum computing is expected to play a pivotal role in next-generation image analysis, offering new possibilities for noise correction and quantum-enhanced vision systems.

1.1. Preliminaries

This subsection presents an overview of fundamental quantum computing concepts, including quantum bits, circuits, and entanglement, enabling quantum computers to manipulate image data beyond the capabilities of classical systems.

1.1.1. Qubit

A qubit, or a quantum bit, is a fundamental information processing unit in quantum computers [16]. Physically, a qubit is an energy state of a subatomic particle—such as an electron or photon—realized using various technologies, including trapped ions [43], cold atoms [44], superconducting circuits [45], nitrogen-vacancy centers in diamond [46], and photonic systems [47]. The state of a qubit is represented using the Dirac notation, i.e., Bra-Ket notations [48]. A Qubit can be in the state $|0\rangle$ or $|1\rangle$ or as a linear superposition of these two states, such that $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ where α and β are complex numbers, satisfying the

normalization condition, $|\alpha|^2 + |\beta|^2 = 1$. On measuring the state of a qubit, we get the state $|0\rangle$ with a probability $|\alpha|^2$ and the state $|1\rangle$ with a probability $|\beta|^2$. A single qubit can be represented as a point on the surface of a three-dimensional Bloch sphere, which visually encodes its state within a two-dimensional Hilbert space [16]. The tensor product of qubit states forms the computational basis of two-qubit states. For example, a single qubit state can be represented as column vectors, i.e.,

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ and therefore, a two-qubit state can be expanded in terms of tensor product as } |00\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, |01\rangle$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, |10\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \text{ and } |11\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}. \text{ A multi-qubit system exists in a superposition of } 2^d \text{ states,}$$

where d is the number of qubits in a system. The relative contribution of each state can be any value between 0 and 1, given that the total probability is 1. For example, if we have two qubits, then it can exist in a superposition of 2^2 states, i.e., $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ where the square of the coefficient of each state represents its probability of occurrence, and the sum of the square of the coefficients would be equal to 1. Thus, if the state of the two qubits is represented as $|\psi\rangle = C_{00}|00\rangle + C_{01}|01\rangle + C_{10}|10\rangle + C_{11}|11\rangle$, then $|C_{00}|^2$ corresponds to the probability of getting $|00\rangle$ and likewise. The measurement of the quantum state leads to the collapse of the wave function to one of the operator's eigenstates. Moreover, the superposition of basis states allows one to perform the same operation on all 2^d basis states in a single step. It provides exponential scaling and computing power to quantum computers as we can store and process 2^d basis states as part of a single wave function [16].

1.1.2. Quantum gates

Quantum gates are unitary matrices that operate on the qubit's state to perform computation. Quantum gates can be categorized as single-qubit or multi-qubit gates, where the former operates on one input and output qubit [16], while the latter involves two or more qubits. Further, quantum gates are reversible in nature [49]. They can perform all classical operations, allowing the creation of quantum circuits that mimic classical logic gates [50–53]. Quantum gates can mathematically be represented as $2^d \times 2^d$ matrices for d input logic operations. Some of the frequently used single and multi-qubit quantum gates are described in Table 1.

1.1.3. Entanglement

Entanglement is central to quantum computing and occurs when a group of subatomic particles exhibits quantum correlation such that the state of individual sub-systems cannot be described independently [16]. In a quest to understand the completeness of quantum theory offering counterintuitive statistical predictions to our understanding of the classical world, Albert Einstein and a couple of his colleagues proposed a thought experiment, now known as the EPR paradox [4]. They argued that if we have a pair of entangled particles with total spin zero, kept far

apart from each other in the universe, then by observing one particle to be in spin-up, one can predict the other particle's state to be spin-down in the same axis without performing any measurement on the other particle. Einstein called this a "spooky action at a distance," and this phenomenon was termed as entanglement. Later, John Bell investigated quantum entanglement by deriving the Bell inequality [54], which was experimentally tested by John Clauser and Stuart Freedman [55]. A d -dimensional qubit system, where $d \geq 2$ can be a composite system of d -Hilbert spaces, i.e., can be represented as a convex combination of tensor product of individual subsystems, or it cannot be represented as such. Algebraically, the former describes a separable state ($|\psi_{12}\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$) and the latter an entangled state ($|\psi_{12}\rangle \neq |\psi_1\rangle \otimes |\psi_2\rangle$). We can get other types of states by combining pure and mixed entangled states with varying degrees of entanglement.

- 1) A two-qubit system of pure and maximally entangled states: Maximally entangled states, also known as Bell states [54,56] or EPR states [57].
- 2) A mixed entangled state: Mixed entangled states such as the Horodecki state [58,59] or Werner State [60].
- 3) A mixed state of two qubits under global unitary operation when their entanglement is maximized: Maximally Entangled Mixed States [61].
- 4) Moreover, for a three-qubit system, we can have separable states, bi-separable states, or genuine entangled states. The pure entangled states of three-qubit systems are classified into two inequivalent classes: the GHZ class [62] and the W-class [63].

One can mathematically quantify the degree of entanglement and determine whether a given state is separable or entangled for the d -qubit system [64].

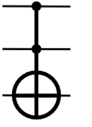
1.2. Quantum image processing

Quantum image processing, an emerging branch of quantum computing, implements strategies by exploiting the quantum hardware for representing, storing, processing, and securing quantum images in various compositions for purposes numerous in quantity [65]. It has the potential to be implemented across multiple research avenues and interdisciplinary domains with practical implications, such as image representation, segmentation, pattern recognition, image data searching, edge and feature extraction, image security, and so on [5]. Some prominent and novel quantum image processing research applications include quantum Fourier [66] and Wavelet transform [67], Grover's quantum image searching [68], quantum machine and deep learning [69], etc. The fundamental attributes of quantum computing, notably entanglement and super positional parallel execution [3–5], open new dimensions for addressing a wide range of challenges, including computational time, memory usage, efficiency, and security. Encoding images in a mechanical composition of any possible quantum computing hardware is essential to facilitate these operations in significantly less computational time [16].

Quantum image processing predominantly falls into *image representation* and *image processing algorithms* [70], as seen in Fig. 1. The former reflects the images in a quantum context, while the latter processes them through quantum circuits to perform distinct image processing tasks. The former is a carved subsegment of quantum multimedia representation, briefly introduced as follows, including quantum movie and audio representations.

- 1) *Quantum image representation* is inspired by classical image representation wherein the information related to color and position of a $d \times d$ sized quantum image is encoded as follows [71]:

Table 1
Common quantum gates.

Gates	Symbol	Matrix	Type	Properties
Identity		$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	1-qubit	It is basis-independent and does not modify the quantum state.
Hadamard		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$		It rotates the states $ 0\rangle$ and $ 1\rangle$ to $ +\rangle$ and $ -\rangle$, respectively, for preparing superposition states.
Pauli-X		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$		It is quantum equivalent to the classical NOT gate, inverses state of $ 0\rangle$ to $ 1\rangle$ and vice versa.
Pauli-Y		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$		It performs rotations by π around the Y-axis of the Bloch sphere.
Pauli-Z		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$		It performs rotations by π around the Z-axis of the Bloch sphere.
C-NOT		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	2-qubit	It flips the target qubit when the control qubit is $ 1\rangle$.
SWAP		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$		It interchanges $ 01\rangle$ to $ 10\rangle$ and vice versa.
Toffoli		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$	3-qubit	<i>CCNOT universal reversible</i> : The third bit is inverted when the first two bits are set to $ 1\rangle$; otherwise, all three bits remain the same.

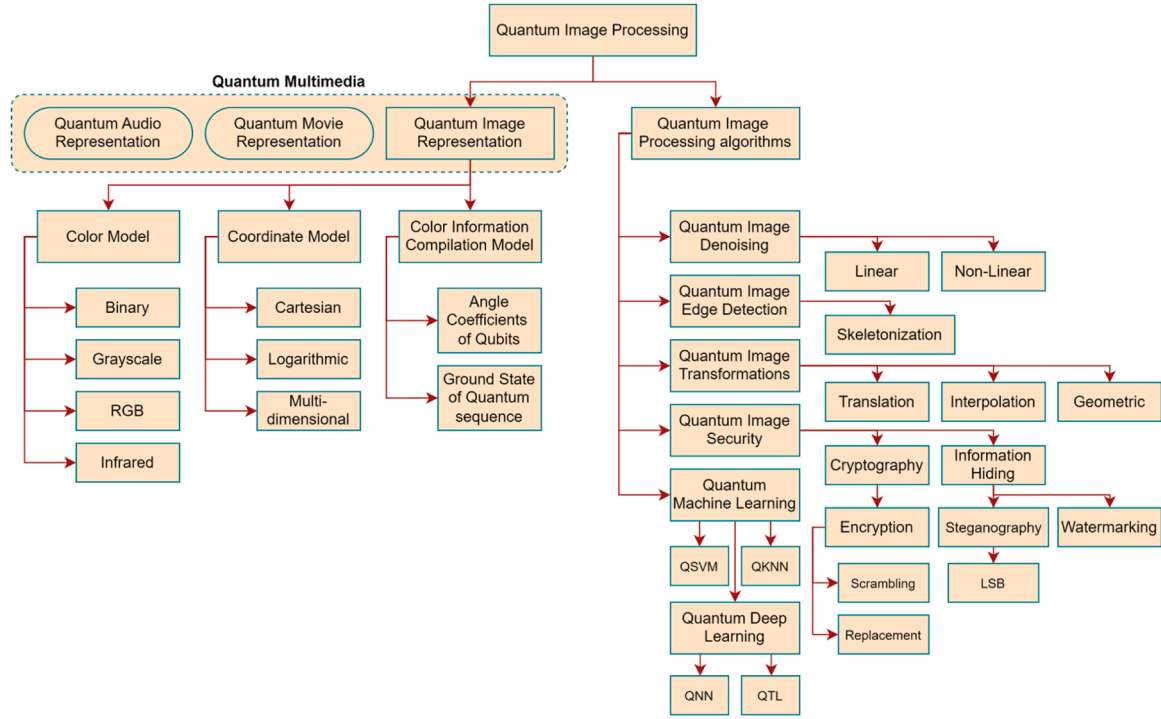


Fig. 1. Taxonomy of quantum image processing, detailing multimedia representation through color and coordinate models, key processing algorithms, security techniques, and quantum machine/deep learning, emphasizing the capabilities of quantum image processing.

$$|\psi_{img}\rangle = \frac{1}{\sqrt{d}} \sum_{h=0}^{d-1} |K_h\rangle \otimes |H\rangle \quad (1)$$

where, $|K_h\rangle$ represents the color information and $|H\rangle$ represents the corresponding position information, while the information related to an image is integrated into a normalized state using the tensor product ($\otimes$) of the two. Furthermore, as discussed later, various models have been developed, providing multiple ways to represent images in a quantum scenario [9,72].

2) *Quantum Movie representation* is a series of static images played at a particular frame transition rate (frames/sec). The improved FRQI model [73] proposed by Iliyasu et al. deals with the representation of movies based on a quantum computer to encode a register consisting of a series of images (a.k.a. strip, $J = 2^j$) simultaneously as follows:

$$|\psi_{mov}\rangle = \frac{1}{2^{j/2}} \sum_{u=0}^{J-1} |F_u(i)\rangle \otimes |k\rangle \quad (2)$$

where, $|k\rangle$ denotes the position of an image in a strip, $k = 0, \dots, 2^j - 1$, j denotes the number of qubits to encode images in a movie, and $|F_u(i)\rangle$ represents image based on the FRQI model. However, it could be extended to work with images based on NEQR [19] and MCQI [74] models. Furthermore, various concepts for quantum movie devices are suggested, like quantum CDs, readers, players, etc., to prepare, retrieve, and recover movies. In addition, various researchers have suggested encryption and decryption over quantum movies [75].

3) *Quantum audio representation* has only been studied recently, with a few research outputs published. Its primary attributes include oscillations, phase, wavelength, and frequency, along with the time

factor being important, as is illustrated in sinusoidal form as follows [76]:

$$Z(T) = A \sin(2\pi fT + \phi) \quad (3)$$

where, A represents amplitude, f represents frequency, ϕ represents phase at time $T = 0$, $f = \frac{\omega}{2\pi}$, $\lambda = \frac{v}{f}$, $\omega = 2\pi f$, λ represents wavelength, and v represents the speed of the audio wave. Relying on ϕ , the results shift as follows [77]:

$$\text{If } \begin{cases} \phi > 0; & \text{audio wave shifts by } \phi/\omega \text{ (secs)} \\ \phi < 0; & \text{audio experiences a delay} \\ +ve \text{ shift} & \text{represents an advance} \end{cases} \quad \begin{cases} \phi = 0; & Z(T) = A \sin(\omega T); \end{cases}$$

The study of quantum audio representation is crucial, as it remains largely unexplored despite its potential in applications such as quantum voice communication, audio encryption, music editing, and multimedia steganography. The research area has kick-started after the successful implementation of the QRDA model [78] wherein, unlike image processing, the amplitude and time information of audio is stored using two-qubit sequences ($|0\rangle$ and $|1\rangle$) entangled together. It is carried out by connecting and merging information from two audios into one. Although the model's overall design resembles NEQR [19], it represents each entity distinctly and is constructed as follows:

$$|\psi_{aud1}\rangle = \frac{1}{2^{U/2}} \sum_{T=0}^{U-1} |B_T\rangle \otimes |T\rangle \quad (4)$$

where, $|B_T\rangle = |B_T^0 B_T^1 B_T^2 \dots B_T^{k-1}\rangle$, $B_T^i \in \{0, 1\}$ denotes the amplitude value of the binary sequence with $|T\rangle = |T_0 T_1 T_2 \dots T_{m-1}\rangle$, and $T_k \in \{0, 1\}$ denotes the time value of its corresponding amplitude.

In light of the findings acquired, it is inferred that the use of $(k+u)$ qubits is necessitated for representing audio of length U and amplitude in the 2^k range, and $L = \text{len}(|\psi_{\text{aud1}}\rangle)$ for the 2^u range. On the other hand, the researchers have also worked on compressing the quantum audio that lessens the number of qubits for the smooth execution of the circuit. Furthermore, a similar approach, FRQA [77], along with quantum audio representation, aims to implement add, inverse, reverse, and delay over audio signals. However, two scenarios correspond to amplitude values, S_T :

$$\text{if } \begin{cases} \text{amp} \geq 0; & S_T \text{ is binary sequence of its value itself} \\ \text{amp} < 0; & S_T^0 = 1, S_T \text{ is 2sCompliment of its absolute value} \end{cases}$$

Further, the FRQA model is formulated as follows:

$$|\psi_{\text{aud2}}\rangle = \frac{1}{2^{\frac{u}{2}}} \sum_{T=0}^{2^u-1} |S_T\rangle \otimes |T\rangle \quad (5)$$

where, $|S_T\rangle = |S_T^0 S_T^1 S_T^2 \dots S_T^{k-1}\rangle$ is 2s compliment of each amplitude value, $S_T^j \in \{0, 1\}$, $j = 0, \dots, k-1$, and $T = 0, \dots, 2^u-1$ is the time information.

Critiquing QRDA [78] by FRQA [77], the former works on positive amplitudes that can potentially be error-prone. For example, when added together, two opposite amplitude increases the overall amplitude instead of negating one another. However, the former conducts various operations like merging audio for practical applications in music technology, speech communication (e.g., conference calls), and compressing audio to reduce the requirement of quantum gates [78]. While the latter undertakes similar (like adding two or more signals) and distinct operations (like signal inversion, delay, and reversal for audio cancellation). However, audio effects like decaying and repetitive echo make the last end to be heard first and vice versa [77]. Quantum multimedia research has paid little attention to audio encryption, limiting its role in secure quantum communication. The embedding of audio within quantum images and videos remains unexplored, leaving a critical gap in multimodal quantum steganography. Expanding this field could enable advancements in quantum watermarking, encrypted teleconferencing, and data embedding techniques, further strengthening the role of quantum audio in next-generation multimedia applications.

1.3. Quantum image representation

Quantum image representation is further segregated into 1) color, 2)

coordinate, and 3) color information compilation-based models, each with respective subclasses. The color model further segregates into 1) binary, 2) grayscale, 3) RGB, and 4) Infrared [70,79–81]. The primary objective of these models is to encode color pixels along with their positions in distinct approaches to represent the energy scores with a bunch of quantum computational bases. If the energy is an electromagnetic wave, its frequency is potentially transformed into a quantum computational basis reflecting color information. Meanwhile, if it is infrared, its altered quantum state can be constructed based on the measurement of the intensity of the wave. However, the binary model-based representation is comparably lucid regarding its algorithmic complexation and storage and is applied primarily for image binarization, segmentation, etc. Contrarily, the grayscale-based model possesses significantly more color information. The RGB model represents color images using two sets of quantum computational bases, one for color information while the other utilizes angles by applying the tensor product with position information. The infrared one encodes infrared images. The coordinate model is further classified into 1) Cartesian, 2) Multidimensional, and 3) Logarithmic-polar. The former is the most convenient and can implement geometric transformations like orthogonal, two-point, image flip, etc. The multidimensional describes multidimensional quantum images, whereas the logarithmic one is used for complex transformations like affine, rotation, blooming, axis-symmetric, symmetric, etc. However, the color information compilation-based models further divide into 1) angle coefficients and 2) the sequence of quantum computational bases that use a vector of angles and the ground state of quantum sequence to compile the color information, respectively. However, Table 2 presents an overview of the models underlying quantum image representation, as shown in Fig. 2, followed by a discussion of the whole taxonomical classification in Section 2.

1.4. Quantum image processing models

As the resolution of images is expanding, conventional image processing faces significant challenges, and combining it with quantum computing appears to be a viable remedy [19,99]. The research around quantum image processing has proven highly effective with practical implications. Many algorithms are proposed to handle quantum images for edge detection, searching image data, steganography, watermarking, and so on, as discussed in Section 3. Contrary to classical paradigms, the prime objective is to capture and process more information adequately.

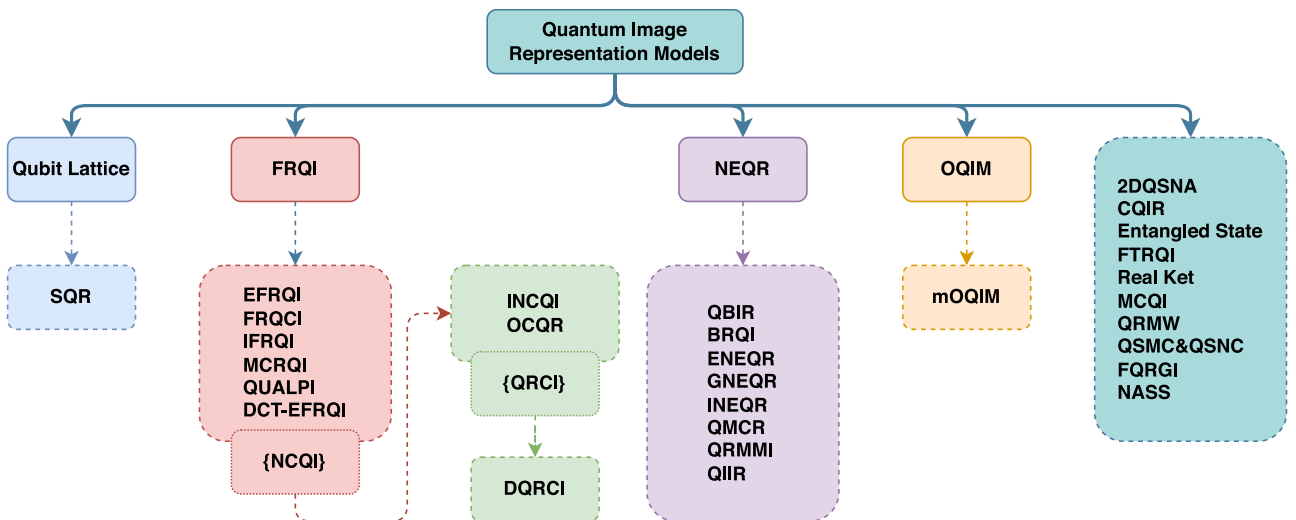


Fig. 2. Taxonomy of quantum image representation models wherein the FRQI [18] and NEQR [19] models have operated as the basis for most research, incorporating improvements into them. These modifications enable the models to behave differently than the base models, making them compatible with various complex operations. However, a few other models are inspired by models like Qubit Lattice [17], QRCI [82], and OQIM [83], whereas multiple models are entirely distinct.

Table 2

* Summary of the quantum image representations models. Models with higher complexity or a greater qubit requirement are underlined, while those with lower complexity and fewer qubits are marked bold. Additionally, models are marked bold if they: support multiple image types, process multiple images simultaneously, operate on quantum movies, or handle both square and rectangular images.

Model [Ref.]	Year	Type of image				No. of Images		Size		Image/ Movie		Retrieval		Complexity	No. of qubits
												<i>D</i>	<i>P</i>		
Qubit Lattice [17]	2003	-	-	✓	-	✓	-	✓	-	✓	-	-	✓	N/A	N/A
Real Ket [84]	2005	-	✓	-	-	✓	-	✓	-	✓	-	-	✓	N/A	N/A
Entangled Image [85]	2010	✓	-	-	-	✓	-	✓	✓	✓	-	✓	-	N/A	N/A
FRQI [18]	2011	-	✓	-	-	✓	-	✓	-	✓	✓	-	✓	$O(2^{4d})$	$2d + 1$
MCRQI [86]	2011	-	-	✓	-	✓	✓	✓	-	✓	✓	-	✓	$O(2^{4d})$	$2d + 3$
NEQR [19]	2013	-	✓	-	-	✓	-	✓	-	✓	-	✓	-	$O(qd2^{2d})$	$2d + q$
QSMC&QSNQ [87]	2013	-	-	✓	-	✓	-	✓	-	✓	-	-	✓	$O(2^{4d})$	$2d + m$
QUALPI [88]	2013	-	✓	-	-	✓	-	✓	✓	✓	-	✓	-	$O(k(d + b)2^{d+b})$	$2d + k$
CQIR [89]	2013	-	✓	-	-	✓	-	✓	-	✓	-	✓	-	$O(\log_2 L.d.2^{2d})$	$2d + j$
MCQI [74]	2013	-	-	✓	-	✓	✓	✓	-	✓	✓	✓	✓	$O(2^{4d})$	$2d + 3$
FQRGI [23]	2013	-	✓	-	-	✓	-	✓	-	✓	-	-	-	N/A	N/A
NASS [90]	2014	-	✓	✓	-	✓	-	✓	✓	✓	-	-	✓	$O(\log 2^d.2^{2d})$	$d + 1$
SQR [81]	2014	-	-	-	✓	✓	-	✓	-	✓	-	-	✓	$O(2^{2d})$	$2d$
INEQR [91]	2015	-	✓	-	-	✓	-	✓	✓	✓	-	✓	-	$O(q(d_1 + d_2)/2^{d_1+d_2+1})$	$2d + 8$ (gray) $2d + 10$ (rgb)
2DQSNQ [92]	2015	-	✓	-	-	✓	-	✓	✓	✓	-	-	✓	N/A	$d + j$
NCQI [93]	2017	-	-	✓	-	✓	-	✓	-	✓	-	✓	-	$O(3k + 2d + 6kd 2^{2d})$	$2d + 3k$
QMCQ [94]	2017	-	-	✓	-	✓	-	✓	-	✓	-	✓	-	$O(3kd \times 2^{2d})$	$2d + 3k$
GNEQR [95]	2018	-	✓	-	-	✓	-	-	✓	✓	-	✓	-	$O(md2^d)$	$d + 8$ (gray) $d + 10$ (rgb)
FRQCI [96]	2018	-	✓	-	-	✓	-	✓	-	✓	-	✓	✓	$O(2^{4d})$	$2d + 1$
BRQI [97]	2018	✓	-	-	-	✓	-	-	✓	✓	-	✓	-	$O(md2^d)$	$2d + 4$ (gray) $2d + 6$ (rgb)
QRMMI [98]	2018	-	-	✓	-	✓	-	✓	-	✓	-	✓	-	$O(d2^{3d})$	$2d + t + q$
QRMW [99]	2018	-	-	✓	-	✓	-	✓	✓	✓	-	✓	-	$O(q2^{(2+2d)})$	$2d + 2 + q$
OCQR [100]	2018	-	-	✓	-	✓	-	✓	-	✓	-	✓	-	$O(3k(2d + 2)2^{2d})$	$2d + k + 2$
QRCI [82]	2019	-	-	✓	-	✓	-	✓	-	✓	-	✓	-	$O(K2^{2d})$	$2d + 3$
IFRQI [101]	2019	-	✓	-	-	✓	-	✓	-	✓	-	-	✓	$O(pd2^{2d})$	$2d + p$
QBIR [102]	2019	-	✓	-	-	✓	-	✓	-	✓	-	✓	-	$O(kd2^{2d})$	$2d + k$
OQIM [83]	2019	-	✓	-	-	✓	-	✓	-	✓	-	-	✓	$O(2^{4d})$	$2d + 2$
mOQIM [83]	2019	-	✓	-	-	-	✓	✓	-	✓	-	-	✓	$O(2^{d+2q})$	$2d + q + 2$
QIIR [103]	2020	-	-	✓	-	✓	-	✓	-	✓	-	✓	-	$O(kd2^{d+b} + 2^k(ck/2))$	$2d + 2k + c$
DQRCI [104]	2020	-	-	✓	-	✓	✓	✓	-	✓	-	✓	-	$O(L2^{2d})$	$2d + 9$
FTQR [105]	2020	-	-	✓	-	✓	-	✓	✓	✓	-	-	✓	N/A	$d + b$ (gray)
INCQI [106]	2021	-	-	✓	-	✓	-	✓	-	✓	-	-	✓	$O(4s + d + b + 4s(d + b) 2^{d+b})$	$d \pm b \pm 4s$
EFRQI [107]	2021	-	✓	-	-	✓	-	✓	-	✓	-	✓	-	$O(2d2^{2d})$	$2d + 1$
ENEQR [107]	2021	-	✓	-	-	✓	-	✓	-	✓	-	-	✓	$O(2d2^{2d})$	$2d + b + 1$
DCT-FRQI [8]	2023	-	✓	-	-	✓	-	✓	✓	✓	-	-	-	N/A	$k + 2d + 1$

* The types of quantum images include binary () , grayscale () , RGB () , and Infrared () . The models either use a single () or multiple () images simultaneously with a size of $2^d \times 2^d$ () square) or $2^d \times 2^b$ () rectangle). All models are compatible with quantum images () ; however, a few are with quantum movies () . Additionally, the models operate probabilistically (*P*) or deterministically (*D*).

In edge detection, the classical ones need to compute the intensity gradient of each pixel of an image; for example, a $2^d \times 2^d$ image necessitates a complexity of $O(2^{2d})$ that is practically infeasible to accomplish. Analogically, quantum computing relying on super positional parallelism and entanglement accomplishes this problem within a minimal computational time. Ordinarily, image processing entails three steps that follow:

- 1) *Preparation*: Transforming classical information (image) into quantum context.
- 2) *Algorithm*: Processing images employing various quantum circuits.
- 3) *Measurement*: Obtaining a traditional image from a quantum system.

1.5. Research methodology

This study explores quantum image processing and derives key observations and critical knowledge by addressing research questions that define the novelty and necessity of the study. This section, as seen in

Fig. 3, unfolds in three phases:

- 1) *Planning* involves clarifying research motivations, formulating research questions, and establishing review protocols.
- 2) *Conducting* entails studying and extracting data from primary sources and synthesizing the information accordingly.
- 3) *Documenting* analyzes and documents critical observations, identifies limitations and research gaps, and suggests future research implications.

Accordingly, the study answers the following questions:

1. *Methodologies for representation*: What methodologies exist in literature to represent classical images in a quantum scenario?
2. *Applications of quantum algorithms*: What are the various applications of quantum image processing in the existing literature?

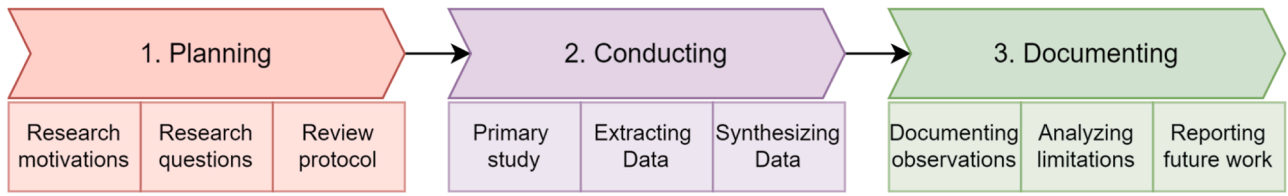


Fig. 3. Overview of the research methodology.

3. *Implementation challenges*: What are the key challenges and hurdles in implementing quantum image processing algorithms for various tasks?
4. *Research gaps*: What are the research gaps in the existing quantum image processing?
5. *Future directions*: What are the potential avenues for future work in quantum image processing?

1.5.1. Literature search

We conducted an extensive literature search across Springer Link, IEEE Xplore, Elsevier (including ScienceDirect), Scopus, the ACM Digital Library, arXiv (including preprints), as well as other peer-reviewed journals indexed in SCI/SCIE, covering 2003 and beyond to capture foundational and recent advancements. To refine the search, a combination of Boolean operators and targeted keywords was employed: ("Quantum" AND "Image") AND ("Processing" OR "Representation" OR "Denoising" OR "Edge Detection" OR "Transformations" OR "Security" OR ("Machine" OR "Deep") AND "Learning"). Initially, 209 papers were retrieved and screened (Fig. 4) to ensure relevance and quality.

1.5.2. Screening process

A screening process was conducted using a three-step approach to refine the selection of relevant literature:

- 1) *Preliminary Filtering*: Initially, 29 duplicate papers and those published before 2003 were excluded. It excluded papers that were available as preprints but later published in reputed journals and conferences.
- 2) *Eligibility Assessment*: Then, articles were screened based on titles and abstracts that excluded 25 irrelevant studies lacking direct relevance to the selected keywords. The remaining articles underwent a full-text review and were assessed based on the following inclusion and exclusion criteria:

- a) *Inclusion Criteria*: 1) Peer-reviewed journal and conference papers from reputed sources written in English. 2) Research covering any of the selected keywords. 3) Studies proposing novel methodologies or presenting experimental results. 4) Articles published beyond 2003.
- b) *Exclusion Criteria*: 1) Non-English publications. 2) Articles published before 2003. 3) Review papers unless they provide a substantial synthesis of relevant research. 4) Studies without empirical or theoretical contributions related to quantum image processing. 5) Project reports and posters.
- 3) *Final Selection*: Ultimately, 135 articles that passed both filtering stages were further analyzed for research significance. This screening ensures that the literature review captures the most relevant advancements in quantum image processing while eliminating redundancy and irrelevant sources.

1.6. Contributions

Nevertheless, this review is the most up-to-date resource offering a comprehensive overview of quantum image processing, encompassing everything from quantum image representation methods to advanced quantum machine and deep learning techniques. However, the main contributions of this study are as follows:

- 1) This study enhances the accessibility of complex concepts while delivering a comprehensive overview of quantum image processing, offering profound insights into its evolution and key advancements beyond 2003.
- 2) It presents an in-depth analysis of quantum image representations, which serve as the foundation for various processing tasks, along with data encoding methods that enable compact data representation for machine learning applications.
- 3) A thorough analysis of real-world applications of quantum image processing algorithms, including quantum image denoising, edge detection, image transformations, and image security, is presented,

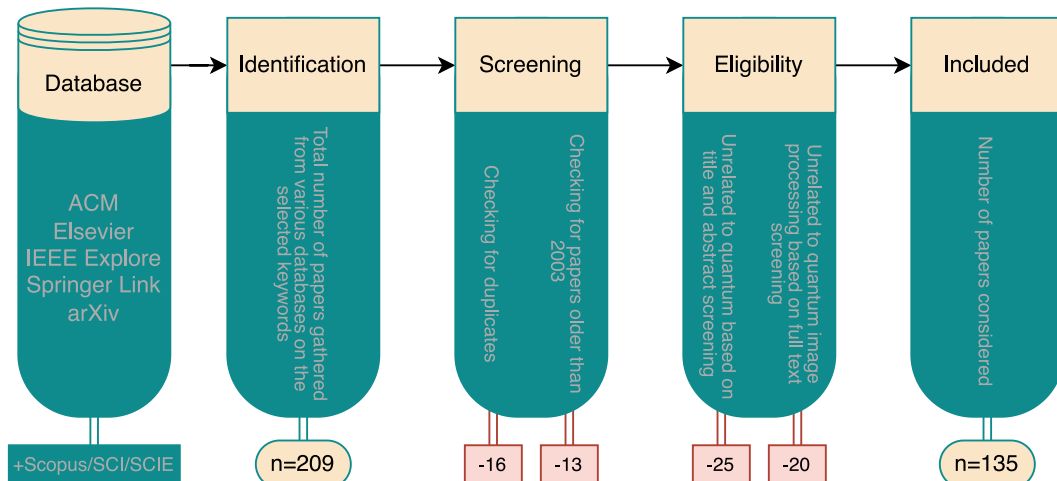


Fig. 4. Workflow of checking the eligibility of research publications.

highlighting their significance, computational advantages, and transformative potential in advancing the field.

- 4) The open areas, research gaps, and future research suggestions presented in this study serve as a guide to exploring unexplored areas and developing novel quantum image processing techniques.

The rest of the paper proceeds as follows: Section 2 provides an extensive survey of quantum image representation models, Section 3 discusses the quantum image processing algorithms and unfolds into various domains, Section 4 presents the open areas and research gaps of quantum image processing and Section 5 concludes the study with significant observations, followed by future perspectives.

2. Quantum image representation models

These models encode and transform classical images into quantum states using the principles of quantum mechanics. Classical image representations, where correlations within an image, if present, are stored in extra space thereby impacting parallel computing [85]. In contrast, quantum image representations use quantum coherence, entanglement, and superposition to achieve remarkable storage capacity while exploiting parallel computing. For instance, FRQI [18] stores a $2^d \times 2^d$ grayscale image using $2d + 1$ qubits, while NASS [90] with only $d + 1$ qubits, significantly reducing resource requirements. As the foundation of quantum image processing, they enable advanced processing like image analysis, compression, pattern recognition, encryption, watermarking, etc., offering speed, scalability, and security advantages over classical methods.

2.1. Qubit lattice and its variant

- The prime concern of this research lies in explaining the obsolete color models and representing a color and an entire image into a qubit and its lattice, respectively [17]. The authors also described a probabilistic approach with maximal certainty for retrieving an image, then contrasted it with the existing classical and stochastic ones and surpassed them. Using this model, the j^{th} and k^{th} pixel values can be encoded as follows:

$$|\psi\rangle = \left(\cos\left(\frac{\theta_{j,k}}{2}\right)|0\rangle + e^{i\varphi} \sin\left(\frac{\theta_{j,k}}{2}\right)|1\rangle \right) \quad (6)$$

They also introduced critical features such as secrecy and eavesdropping detection to enhance quantum image security. The former provides a random output to the eavesdropper, while the latter detects it corresponding to the right and wrong bases.

- Inspired by Qubit Lattice [17], the SQR model [81] is based on the physical properties of infrared images and resolves various issues in the former and FRQI-based models. Infrared radiation energy from objects is converted to a quantum state by a converter adept at sensing and recording infrared radiation and generating output as a quantum state. The converter initializes various states for varying energies and is stored in a quantum state as follows:

$$|\psi_{jk}\rangle = \cos\frac{\theta_{jk}}{2}|0\rangle + \sin\frac{\theta_{jk}}{2}|1\rangle \quad (7)$$

where, $\theta_{jk}(j = \{1, \dots, d\}, \theta_{jk} = [0, 2\pi])$ and $k = \{1, \dots, b\}$ are real numbers, d and b denote dimensions of the converter array. It utilizes a lucid function to transform an infrared image into a quantum state vector, subsequently employed for quantum image processing tasks such as compression and filtering. As per the paper's findings [81], it outpaces existing image processing methods considering compression ratio and picture quality. Since it employs two d -qubit states to store minimum and maximum energy values, additional $2d$ quantum gates are necessitated to prepare an infrared quantum image.

2.2. Real ket

- According to this model [84], an image's pixel values can be constituted using a real ket and adequate block-structured addressing wherein the final state is formulated using a matrix product. However, it is synthesized in three operations: 1) an image is cast into a quantum register that relies on local degrees of freedom in a logarithmic fashion wherein entanglement of the state reflects pixel values being correlated to their respective positions, 2) the output of 1) is formulated using matrix product wherein a less entanglement is seen in smooth images, and 3) a truncation scheme for entanglement implies a compression algorithm in a classical setting. The model works on blocks of pixels of an image wherein each block represents a dimension of $2^1 \times 2^1$. However, its general representation of a $2^d \times 2^d$ image is as follows:

$$|\psi_{2^d \times 2^d}\rangle = \sum_{j_{1-4}} K_{jd} |j_d, \dots, j_1\rangle \quad (8)$$

where $K_{jd} = K_{jd}, \dots, K_{j1}$ represent pixel values and j_{1-4} represent four pixels of $2^1 \times 2^1$ block [84].

2.3. Entangled image

- As many models require auxiliary qubits to hold additional information about correlation, spatial arrangement of pixels, etc., this quantum binary image model compresses them by exploiting entanglement [85]. The model stores, retrieves, and reconstructs binary geometric shapes like triangles, squares, etc. For example, to represent a triangle and a square, three and four qubits, respectively, will be set to $|1\rangle$ while the remaining as $|0\rangle$. These images would be retrieved using Grover's search algorithm [12]. However, if there are two geometric shapes (say two triangles), additional information would still be required to discern which pixel corresponds to which triangle. Nonetheless, entanglement establishes a nonlocal correlation between the vertices containing pixel locations of the same triangle. The memory state for two triangles is formulated as follows:

$$|\psi_{\Delta}\rangle = \otimes_{j=1}^d |0\rangle_j \otimes |GHZ\rangle_{lmn} \otimes |GHZ\rangle_{opq} \quad (9)$$

where, $j \neq l, m, n, o, p, q$ and $|GHZ\rangle = (|000\rangle + |111\rangle)/\sqrt{2}$, vertices $\{l, m, n\}$ and $\{o, p, q\}$ belong to triangles Δ_1 and Δ_2 , respectively.

2.4. FRQI and its variants

- The widely used model, FRQI [18], captures both color and position information of each pixel using a vector of angles in normalized quantum states, formulated as follows:

$$|\psi(\theta)\rangle = \frac{1}{2^d} \sum_{k=0}^{2^d-1} \underbrace{(\cos(\theta_k)|0\rangle + \sin(\theta_k)|1\rangle)}_{\text{color}} \otimes \underbrace{|k\rangle}_{\text{position}} \quad (10)$$

where, $\theta_k \in [0, \frac{\pi}{2}]$ and $k = 0 \dots 2^d - 1$. It is achieved by applying (Fig. 5) 1) superposition by Hadamard on each qubit followed by 2) CNOT and 3) phase shift operators to every standard basis to construct a joint state as below:

$$R|H\rangle = \left(\prod_{k=0}^{2^d-1} K_k \right) |H\rangle = |I(\theta)\rangle \quad (11)$$

where K_k denotes rotation of pixels. Hence, we can potentially encode the intensities by rotating these standard bases. The unitary operators transform the initial state into the FRQI state. Based on the FRQI model [18], advanced image processing algorithms like GTQI [108], rGTQI [109,110], and CTQI [111] manipulate pixel positions and color information for applications like encryption.

- Nonetheless, the shortcoming of FRQI [18], the inability to process

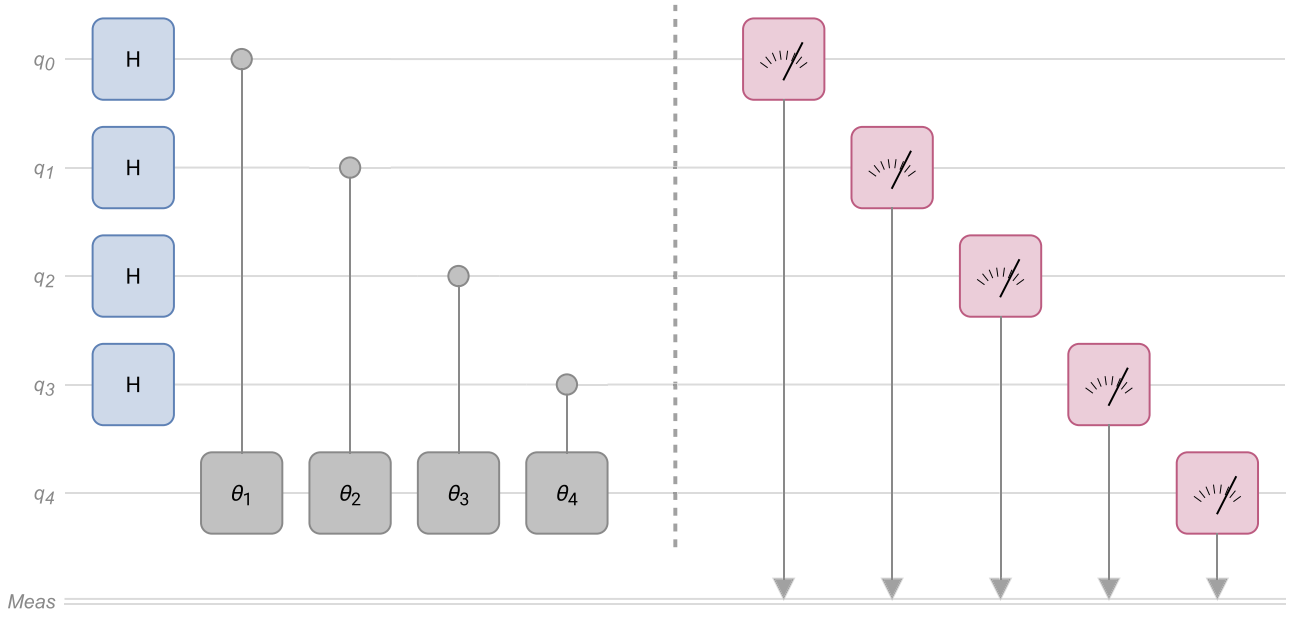


Fig. 5. FRQI quantum circuit for a 2×2 grayscale image, where one qubit encodes the color information of four pixels, and four qubits represent pixel positions. The entire image is encoded using the tensor product of these qubits.

RGB color images, is resolved by FRQCI [96], wherein phase variables ϑ and θ are used to store R and GB colors, respectively. The model is constructed as follows wherein $|k\rangle$ represents a 2^{2d} dimensional computational basis:

$$|\psi(\theta, \vartheta)\rangle = \frac{1}{2^d} \sum_{k=0}^{2^{2d}-1} |k\rangle \underbrace{|c_k\rangle}_{\cos \frac{\theta_k}{2} |0\rangle + e^{i\vartheta_k} \sin \frac{\theta_k}{2} |1\rangle} \quad (12)$$

where, $\theta_k = \frac{c_k^R \times \pi}{2^{2d}-1}$, $\vartheta_k = \frac{(c_k^G \times 256 + c_k^B) \times 2\pi}{256 \times 2^{2d}-1}$, c_k^R , c_k^G , and c_k^B represent RGB color values of the k th pixel. Further, various color and geometric transformations and image encryption are ensured using two phases: 1) scrambling positions and 2) transforming colors followed by effective decryption.

• Similarly, MCQI [74] simultaneously encodes RGB channels using a few normalized qubits that can be used for watermarking, quantum movies, and image data searches. While it uses MC-PPT [74] to shift angle vectors to three, its time complexity is the same as FRQI [18] but with a different approach. The MCQI representation is as follows:

$$|\psi(d)\rangle = \frac{1}{2^{d+1}} \sum_{j=0}^{2^{2d}-1} (|C_{rgb}^j\rangle \otimes |j\rangle) \quad (13)$$

$$\begin{aligned} |C_{rgb}^j\rangle = & \cos\theta_R^j |000\rangle + \cos\theta_G^j |001\rangle + \cos\theta_B^j |010\rangle + \sin\theta_R^j |100\rangle \\ & + \sin\theta_G^j |101\rangle + \sin\theta_B^j |110\rangle + \cos 0 |011\rangle + \sin 0 |111\rangle \end{aligned} \quad (14)$$

where, $\{\theta_{R_1}^j, \theta_{G_2}^j, \theta_{B_3}^j\} \in \{0, \frac{\pi}{2}\}$ denote 3 encoding angles for RGB channels considering the j th pixel. $|C_{rgb}^j\rangle$ and $|j\rangle$ encode color and position information, respectively. Since six coefficients are enough to encode all the information about the image, two additional ones ($\cos 0$ and $\sin 0$) are set to constant to indicate no information. Furthermore, a

strip representation [73] is applicable to encode 2^d FRQI images with fewer quantum resources offering a potential approach for creating quantum movies.

• However, MCRQI [86] advances different image processing tasks, such as swapping channel operations based on FRQI [18]. It is identical to the MCQI [74]; instead of solely employing RGB channels, it uses an additional α channel for transparency:

$$|\psi(d)\rangle = \frac{1}{2^{d+1}} \sum_{j=0}^{2^{2d}-1} (|C_{rgba}^j\rangle \otimes |j\rangle) \quad (15)$$

$$\begin{aligned} |C_{rgba}^j\rangle = & \cos\theta_r^j |000\rangle + \cos\theta_g^j |001\rangle + \cos\theta_b^j |010\rangle + \cos\theta_a^j |011\rangle \\ & + \sin\theta_r^j |100\rangle + \sin\theta_g^j |101\rangle + \sin\theta_b^j |110\rangle + \sin\theta_a^j |111\rangle \end{aligned} \quad (16)$$

It is stated that when coupled with other models [108–110], it can potentially be used for quantum watermarking and movie representation. An enhanced variant of MCQI [74], the CFQM [112] framework for quantum movies, has also been presented. • Furthermore, intending to work more compatibly with grayscale images, the FQRGI model [23] is formulated as follows:

$$|\psi(\theta)\rangle = \frac{1}{2^d} \sum_{k=0}^{2^{2d}-1} |G_k\rangle \otimes |k\rangle \quad (17)$$

$$|G_k\rangle = (|0\rangle + e^{i\theta_k} |1\rangle) \quad (18)$$

where, $\theta_k \in \{0, \frac{\pi}{2}\}$, $k = 0 \dots 2^{2d} - 1$, $\theta_0 \dots \theta_{2^{2d}-1}$ represent phase vector that holds grayscale information, and $|k\rangle$, wherein $k = 0 \dots 2^{2d} - 1$, denote 2^{2d} dimensional quantum computational basis states, $|G_k\rangle$ and $|k\rangle$ hold grayscale and position information of an image, respectively.

• In addition, to work with quantum images based on logarithmic polar coordinates, the QUALPI model [88] stores and processes them for transformations like affine, symmetry, rotation, etc. The model is formulated as follows:

$$|\psi\rangle = \frac{1}{\sqrt{2^{u+k}}} \sum_{\rho=0}^{2^u-1} \sum_{\theta=0}^{2^k-1} |f(\rho, \theta)\rangle \otimes |\rho\rangle \otimes |\theta\rangle \quad (19)$$

where, $f(\rho, \theta)$ represents the gray score of the current pixel, $|\rho\rangle$ represents the radius of the polar coordinate, and $|\theta\rangle$ denotes the horizontal angle. In addition, an image registration approach, such as brute force, is also presented as being comparably speedier than the traditional ones [88].

2.5. QSMC & QSNIC

• The QSMC & QSNIC model [87] stores images using a set of $(2d+j)$ qubits representing d colors (QSMC) and j coordinates (QSNIC) wherein employing bijective functions, the color and coordinates are described using angles φ and θ , respectively and formulated as follows:

$$|\psi\rangle = \frac{1}{2^d} \sum_{j=0}^{2^d-1} \underbrace{(\cos\varphi_j|0\rangle + \sin\varphi_j|1\rangle)}_{QSMC} \otimes \underbrace{(\cos\theta_j|0\rangle + \sin\theta_j|1\rangle)}_{QSNIC} \quad (20)$$

where, $\varphi_j, \theta_j \in [0, \frac{\pi}{2}]$, and $j = 0, \dots, 2^d - 1$. In addition, quantum image segmentation and compression algorithms are presented, wherein the former necessitates a complexity of $O(j\sqrt{d})$, while staying compatible with gray and color images, the latter is lossless, representing no information is lost during compression.

2.6. NEQR and its variants

• The NEQR [19] model uses entangled two-qubit sequences to store quantum images, incorporating grayscale and positional information for each pixel (Fig. 6). A $2^d \times 2^d$ image is represented by this model as follows:

$$|\psi\rangle = \frac{1}{2^d} \sum_{y=0}^{2^d-1} \sum_{x=0}^{2^d-1} |f(y, x)\rangle |yx\rangle \quad (21)$$

where, $|f(y, x)\rangle$ represents entangled information of color values at $|yx\rangle$ position. Nonetheless, its first step is the same as FRQI [18], but then it stores the grayscale information in 2 to 2^{2d} subphases for each pixel. It is also asserted that the NEQR demands a complexity of $O(qd2^{2d})$ comparably speedier than FRQI [18], but the former uses more qubits

than the latter. Moreover, this model also suggests and performs partial and statistical-based color operations, wherein CNOT and Toffoli gates link the color and position information to easily process any pixel of a quantum image.

• As the name INEQR [91] suggests, it is an improvised version of NEQR [19] that relies on interpolation, wherein new or superfluous pixels are added or eliminated. While interpolation methods, such as nearest neighbor, bicubic, and bilinear, are available, the authors opted for the nearest neighbor approach due to its speed and simplicity, despite its inferior results. As the NEQR [19] considers $2^d \times 2^d$ images, scaling the image results in distinct rows and columns. To resolve this issue, the INEQR model [91] is compatible with $2^d \times 2^b$ images and is constructed as follows:

$$|\psi\rangle = \frac{1}{2^{\frac{d+b}{2}}} \sum_{y=0}^{2^d-1} \sum_{x=0}^{2^b-1} \bigotimes_{i=0}^{q-1} |C_{yx}^i\rangle |yx\rangle \quad (22)$$

where, $|yx\rangle = |y\rangle|x\rangle = |y_0y_1y_2\dots y_{d-1}\rangle|x_0x_1x_2\dots x_{b-1}\rangle$, $|C_{yx}^i\rangle = f(y, x)$ represents the binary sequence encoding the grayscale value of the pixel (Y, X) , and $y_i, x_i \in \{0, 1\}$. The model uses y (d -qubits), x (b -qubits), and $f(y, x)$ (q -qubits) for Y, X coordinates and grayscale intensities of a quantum image, respectively. Further, this model allows for the upscaling and downscaling of a quantum image through quantum circuits [91].

Another improvised version of NEQR [19] stores signals of 2D integers followed by its circuit design; however, the design is the same in both [95].

• Further, a $2^d \times 2^d$ image based on GNEQR [95] requires $d+8$ and $d+10$ qubits to store color information of grayscale and color images, leading to increased storage requirements.

• Furthermore, the QMCR model [94] improves NEQR [19] by adapting it for color images. It encodes an image using two sequences of entangled qubits, one for position information and the other for color values of a pixel. If 2^b is the grayscale range of each channel, then a $2^d \times 2^d$ image can be represented as follows:

$$|\psi\rangle = \frac{1}{2^d} \sum_{j=0}^{2^d-1} \sum_{k=0}^{2^d-1} |C_{RGBjk}\rangle \otimes |jk\rangle \quad (23)$$

where, $|C_{RGBjk}\rangle$ represents state to encode information of RGB color channels, considering the jk pixel and

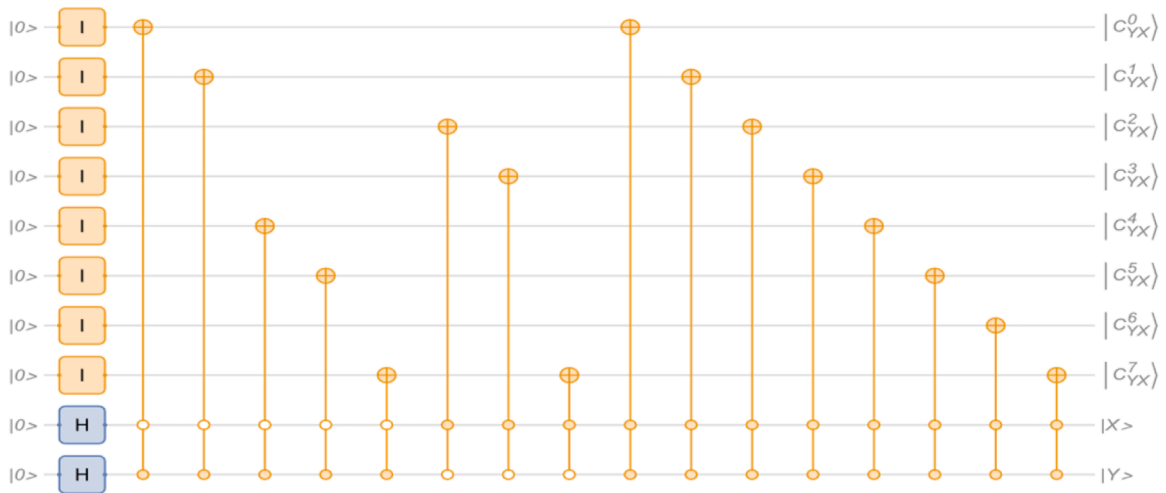


Fig. 6. The NEQR quantum circuit [19] for a 2×2 image that utilizes entangled two-qubit sequences to encode grayscale and positional information for each pixel. Unlike FRQI, NEQR stores grayscale values in multiple subphases, enabling efficient color operations while requiring more qubits.

$|c_{rgbjk}\rangle = |R_{jk}\rangle \overbrace{|G_{jk}\rangle \text{UNDERSTRETCHYTRUE}}^{\text{green}} \overbrace{|B_{jk}\rangle \text{UNDERSTRETCHYTRUE}}^{\text{red}}$ represent three color channels encoded, $|R_{jk}\rangle = |r_{jk}^{b-1} \dots r_{jk}^0\rangle$, $|G_{jk}\rangle = |g_{jk}^{b-1} \dots g_{jk}^0\rangle$, and $|B_{jk}\rangle = |b_{jk}^{b-1} \dots b_{jk}^0\rangle$, R_{jk} , G_{jk} , and $B_{jk} \in \{0, \dots, 2^b - 1\}$ and r_{jk}^m, g_{jk}^m , and $b_{jk}^m \in \{0, 1\}$ wherein the model necessitates $3b + 2d$ qubits to represent the quantum image with a complexity of $O(3bd \times 2^{2d})$. The image preparation involves two main steps: 1) $3b$ and $2d$ qubits represent the grayscale value and pixel location, respectively, initialized to $|0\rangle$, followed by applying Hadamard and Identity transformations. 2) For the RGB channels, the grayscale value is set for each pixel using $2d$ -CNOT gates. Additional operations include resetting to grayscale, channel swapping, and color transformations of specific pixels [94].

• The QRMMI model [98], an enhanced version of NEQR [19], addresses the issue of simultaneously representing and processing multiple images. The method is based on mapping the pixels of various images into the superposition of two entangled quantum state sequences, wherein the initial sequence holds the color information while the other deals with position and sequence numbers to mark multiple images. It uses a quantum random mapping function to generate a series of random numbers, which are then used to map the pixels of the input images into the Hilbert space. If this model is to illustrate N images of $2^d \times 2^d$ size alongside a color depth of k , subsequently, the color content ($f_c(Y, X)$) of pixel (Y, X) in the c^{th} image is likely to be formulated with a binary sequence. However, for representing N images, the model is described as follows [98]:

$$|\psi\rangle = \frac{1}{\sqrt{2^{2d+n}}} \sum_{d=0}^{2^n-1} \sum_{Y=0}^{2^d-1} \sum_{X=0}^{2^d-1} \underbrace{|C_{cYX}^{k-1} \dots C_{cYX}^0\rangle}_{|C_{f_c(Y,X)}\rangle} \otimes |CYX\rangle \quad (24)$$

where, $C_{cYX}^k \in \{0, 1\}$, $f_c(Y, X) \in \{0, 2^k - 1\}$, $|CYX\rangle = |c_{n-1} \dots c_0\rangle |y_{n-1} \dots y_0\rangle |x_{n-1} \dots x_0\rangle$, $|C\rangle$ represents sequence number and $|YX\rangle$ represents location information of C^{th} image. Further, $|C_{f_c(Y,X)}\rangle$ represents the color information of each pixel (Y, X) , and $n = \text{ceil}(\log_2 N)$. The QRMMI model [98] includes a 3D Arnold-transform and a scaled Zhongtang chaotic system to encode the quantum state. These procedures add randomness to the state, making deciphering challenging without a proper decryption key. It addresses the challenges of encrypting multiple images simultaneously, such as data interference and loss of image information [98].

• Although a few models work with the representation of quantum block images, the QBIR model [102] tackles it with the assistance of the NEQR model. Two qubit sequences entangled together encode grayscale values and the position of image blocks, respectively. This model represents a $2^d \times 2^d$ image with $2^b \times 2^b$ blocks as:

$$|\psi\rangle = \frac{1}{2^d} \sum_{j=0}^{2^b-1} \sum_{l=0}^{2^b-1} \sum_{y=0}^{2-b} \sum_{x=0}^{d-b} \underbrace{|C_{jlyx}^{k-1} \dots C_{jlyx}^0\rangle}_{|C(j, l, y, x)\rangle} \otimes \underbrace{|jl\rangle \otimes |yx\rangle}_{|jlyx\rangle} \quad (25)$$

where, $|jlyx\rangle$ represents position information, wherein, $|jl\rangle$ is the location of blocks and $|yx\rangle$ is the position of pixels in each block. However, $2d + k$ qubits are required to represent an image with the aforementioned size. Furthermore, similar to NEQR are the steps: 1) accumulating information related to blocks and corresponding pixels by employing a single I and two H gates for a superposition state, evolving into an intermediate state, 2) setting each pixel's (x, y) gray score in the (j, l) block. However, it works on a single pixel while the others follow the same route;

hence, 2^{2d} suboperations are required.

• The traditional representation of indexed images uses two data structures: the data and palette matrices. The pixel's color is determined by using the data matrix value as an index to the palette matrix. Inspired by it, a quantum variant is presented by QIIR [103] that considers two data structures, viz. $|\psi_D\rangle$ and $|\psi_P\rangle$ to address quantum data and palette matrices, respectively. For a $2^d \times 2^d$ image with k -bit depth, these can be calculated, respectively, as follows:

$$|\psi_D\rangle = \frac{1}{2^d} \sum_{Y=0}^{2^d-1} \sum_{X=0}^{2^d-1} |I_{YX}\rangle \otimes |YX\rangle \quad (26)$$

$$|\psi_P\rangle = \frac{1}{\sqrt{2^k}} \sum_{l=0}^{2^k-1} |C_l\rangle \otimes |l\rangle \quad (27)$$

where, $|YX\rangle = Y_{d-1} \dots Y_0 X_{d-1} \dots X_0$ denotes 2D position information for every pixel value ($|I_{YX}\rangle = |I_{YX}^{k-1} \dots I_{YX}^0\rangle$) that also is an index of the palette matrix. Further, $|l\rangle = |l_{k-1} \dots l_0\rangle$ is the index of the palette matrix and $|C_l\rangle$ represents the red, green, and blue values of a single color. Nonetheless, $2d + k$ and $24 + k$ qubits are required to represent matrices of data and palette, respectively, whereas a classical approach requires $2^d \times 2^d \times k$ and $2^k \times 24$ classical bits, respectively. In addition, various image processing operations have also been suggested, like rotating orthogonally, cyclic shift, color lookup, inversion, and replacement, followed by steganography and its successful message extraction.

2.7. 2DQSN

• The 2DQSN model [92] addresses the limitations of FRQI [18] and NEQR [19], which work only with square ($2^d \times 2^d$) images and require $2d + j$ qubits. Instead, 2DQSN [92] can handle both square and rectangular ($2^d \times 2^b$) images using only $d + b$ qubits. It uses two features: 1) A dual representation of 2D images with row and column vectors of d and b dimensions, respectively, constructed using tensor products:

$$|J\rangle_u = |j\rangle^{\otimes d} \quad (28)$$

and

$$\langle K|_s = \langle k|^{s \otimes b} \quad (29)$$

where, $j \in \{0, 1\}$, $u \in \{1, D\}$, $k \in \{0, 1\}$, $s \in \{1, B\}$, $d = \log_2(D)$, $b = \log_2(B)$, $|J\rangle_u$ and $\langle K|_s$ represent quantum states at u^{th} row and s^{th} column, respectively, forming the pixel location:

$$L_{u,s} = |J\rangle_u \otimes \langle K|_s \quad (30)$$

2) Representing pixel amplitudes with scalars (α) for the corresponding 2D basis, the whole image is represented as [92]:

$$I = \sum_{u=1}^D \sum_{s=1}^B \alpha_{u,s} L_{u,s} \quad (31)$$

2.8. CQIR

• Using qutrits, the CQIR model [89] represents and processes quantum images based on ternary logic, then constructs circuits for operations like complement, binarization, and histogram equalization of images. Nevertheless, due to its incompetence in manipulating color images, it is only interoperable with grayscale images. It is formulated as follows:

$$|\psi\rangle = |C\rangle_b \times |P\rangle_{2d} = \frac{1}{2^d} \sum_{j=0}^{2^d-1} \sum_{k=0}^{2^b-1} \alpha_{jk} |k\rangle |j\rangle \quad (32)$$

where, $|P\rangle_{2d}$ and $|C\rangle_b$ represent registers to store pixel and color information in $2d$ and b qubits, respectively, α_{jk} represents the color of the pixel at j th location.

2.9. NCQI and its variants

- An improvised NCQI color model [93] uses additional auxiliary qubits to handle noise and perform geometric operations (like flipping

$$|\psi\rangle = \frac{1}{2^{d+1}} \sum_{y=0}^{2^d-1} \sum_{x=0}^{2^d-1} \left(|R_{yx}\rangle \otimes \underbrace{|00\rangle}_{r_ci} + |G_{yx}\rangle \otimes \underbrace{|01\rangle}_{g_ci} + |B_{yx}\rangle \otimes \underbrace{|10\rangle}_{b_ci} + |U_{yx}\rangle \otimes \underbrace{|11\rangle}_{s_ci} \right) |yx\rangle \quad (36)$$

and rotating) and other tasks (like color transformation, compression, feature extraction, etc.) on quantum images. It supports quantum gates like CNOT, SWAP, and σ_x for operations on the RGB and α channels. This model overcomes the limitations of NCQI [93], such as restricted image dimensions and transformation methods. The NCQI model [93] is formulated as:

$$|\psi_{NCQI}\rangle = \frac{1}{2^d} \sum_{y=0}^{2^d-1} \sum_{x=0}^{2^d-1} |f(y, x)\rangle \otimes |yx\rangle \quad (33)$$

- However, the improvised NCQI model, INCQI [106], is represented as follows:

$$|\psi_{INCQI}\rangle = \frac{1}{2^{\frac{d_1+d_2}{2}}} \sum_{j=0}^{2^{d_1}-1} \sum_{k=0}^{2^{d_2}-1} |c(j, k)\rangle \otimes |jk\rangle \quad (34)$$

Where, $|f(y, x)\rangle = \underbrace{|R_{s-1} \dots R_0\rangle}_{red} \underbrace{|G_{s-1} \dots G_0 B_{s-1} \dots B_0\rangle}_{green \quad blue}$ is a binary sequence

that represents the color values of the respective pixel at (y, x) position, $|c(j, k)\rangle = \underbrace{|R_{s-1} \dots R_0\rangle}_{red} \underbrace{|G_{s-1} \dots G_0 B_{s-1} \dots B_0 \alpha_{s-1} \dots \alpha_0\rangle}_{green \quad blue \quad \alpha}$ is also for the same but

with an auxiliary quantum bit (α) to acquire an intermediate score for each pixel after transformation. The results achieved by INCQI were better than NCQI; however, the practical experiments were only tested on 2×4 grayscale images due to the limitations in the number of qubits in IBMQ through Qiskit. This is why attempting to work on d -dimensional (more specifically when $d \gg \gg$) is practically

challenging [106].

- Further, the OCQR [100] provides an optimum quantum representation for color images where a single qubit represents each pixel's color. The model uses a 3D sequence to store quantum images, including the value, channel index, and position information. This model is based on NCQI [93] and is described as follows:

$$|\psi\rangle = \frac{1}{2^{d+1}} \sum_{y=0}^{2^d-1} \sum_{x=0}^{2^d-1} |c(x, y)\rangle |ci\rangle |yx\rangle \quad (35)$$

where, $|R_{yx}\rangle = |R_{yx}^{k-1} \dots R_{yx}^0\rangle$, $|G_{yx}\rangle = |G_{yx}^{k-1} \dots G_{yx}^0\rangle$, $|B_{yx}\rangle = |B_{yx}^{k-1} \dots B_{yx}^0\rangle$, and $|U_{yx}\rangle = |U_{yx}^{k-1} \dots U_{yx}^0\rangle$, ci denotes the index of the channel, r_ci , g_ci , b_ci , and s_ci denote the index of red, green, blue, and spare channels, respectively. However, s_ci has not been utilized and is suggested for future research to address the transparency of each pixel. Furthermore, it necessitates k , 2, and $2d$ qubits to encode channel value, index, and position information, respectively, which results in the utilization of $2d+k+2$ qubits. Contrasting NCQI, this model requires fewer qubits to hold an image and pixel values [100].

- Another improved NCQI model, the QRCI model [82], stores and processes quantum color images by relying on the color and bitplanes. It employs two entangled qubit sequences: the first sequence stores the color values of RGB channels, while the other encompasses the corresponding position and bitplane information (Fig. 7). To represent a $2^d \times 2^d$ image, the QRCI model [82] is formulated as follows:

$$|\psi\rangle = \frac{1}{\sqrt{2^{2d+3}}} \sum_{L=0}^{2^3-1} \sum_{Y=0}^{2^d-1} \sum_{X=0}^{2^d-1} |C_L(Y, X)\rangle \otimes |LYX\rangle \quad (37)$$

where, $|C_L(Y, X)\rangle = |R_{LYX} G_{LYX} B_{LYX}\rangle$ represents color information of RGB channels of (Y, X) pixel in the L^{th} bitplane. However, it is carried out in these steps: 1) a quantum state is initialized and set to $|0\rangle$, 2) employing three and $2d+3$ quantum I and H gates, respectively, the position and bitplane information is accumulated to generate an intermediate state, 3) the color information for each pixel in 8-bit planes is stored wherein quantum oracle is used to store binary information of each channel.

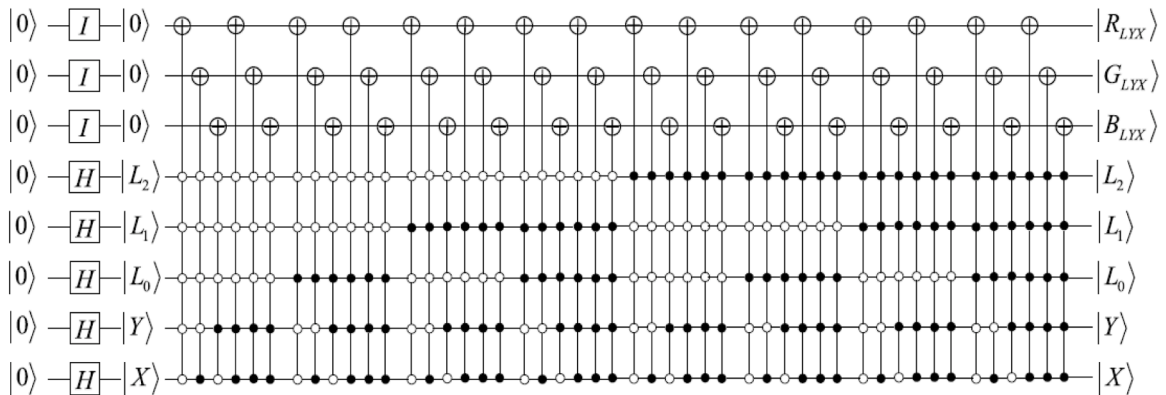


Fig. 7. Quantum circuit of QRCI [82] for a 2×2 -color image. It requires $2d+6$ qubits for $2^d \times 2^d$ color image making it 2^{18} times faster than NCQI.

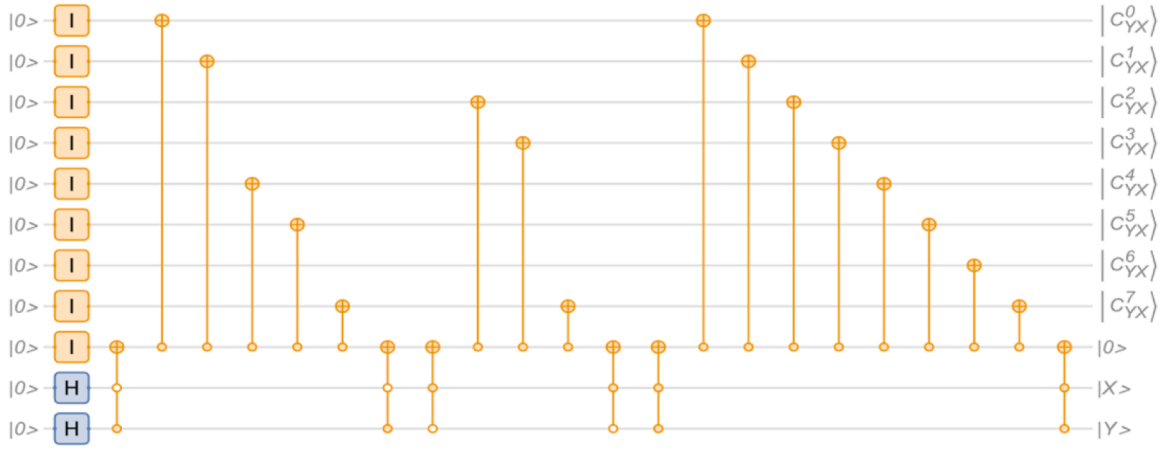


Fig. 8. Quantum circuit for a 2×2 grayscale image using the ENEQR model [107]. The time complexity of ENEQR ($2d2^{2d}$) is lower than that of NEQR ($qd2^{2d}$) for a $2^d \times 2^d$ image.

• However, working on two quantum images simultaneously is addressed by the DQRCI model [104], as most approaches consider only a single image. It encodes RGB color channels, bitplanes, and position information using three entangled qubit sequences. To store two $2^d \times 2^d$ quantum color images, it is formulated as follows:

respectively. To represent a $2^d \times 2^d$ image, the EFRQI [107] employs a partial negation operator to store the values of pixels in the amplitudes of qubits using a single qubit entangled with a sequence of qubits associated with its corresponding location. It is formulated as follows:

$$|\psi\rangle = \frac{1}{\sqrt{2^{2d+3}}} \sum_{N=0}^{2^3-1} \sum_{Y=0}^{2^d-1} \sum_{X=0}^{2^d-1} \underbrace{R_{NYX}^o G_{NYX}^o B_{NYX}^o R_{NYX}^t G_{NYX}^t B_{NYX}^t}_{C_N^o(X, Y)} \underbrace{UNDERSTRETCHYTRUE}_{C_N^t(X, Y)} \otimes |N\rangle \otimes |YX\rangle \quad (38)$$

Where, $|N\rangle$ denotes bitplane, $|YX\rangle$ denotes position information, $C_N^o(Y, X)$, and $C_N^t(Y, X)$ represent color information of the first (o) and the second (t) image corresponding to the RGB channels. However, instead of consuming $2(2d+24)$ qubits, it decreases to $2d+9$.

2.10. Other FRQI and NEQR variants

• Two enhanced variants of FRQI [18] and NEQR [19] have been stated that rely on amplitude and basis state representations,

$$|\psi_{EFRQI}\rangle = \frac{1}{2^d} \sum_{Y=0}^{2^d-1} \sum_{X=0}^{2^d-1} (a_{YX}|0\rangle + b_{YX}|1\rangle) |YX\rangle \quad (39)$$

where, a and b are the amplitudes of the states.

• Likewise, the following ENEQR [107] representation can be utilized to illustrate a $2^d \times 2^d$ image with a grayscale range of 2^k . It employs two sequences of entangled qubits to hold each pixel's grayscale value

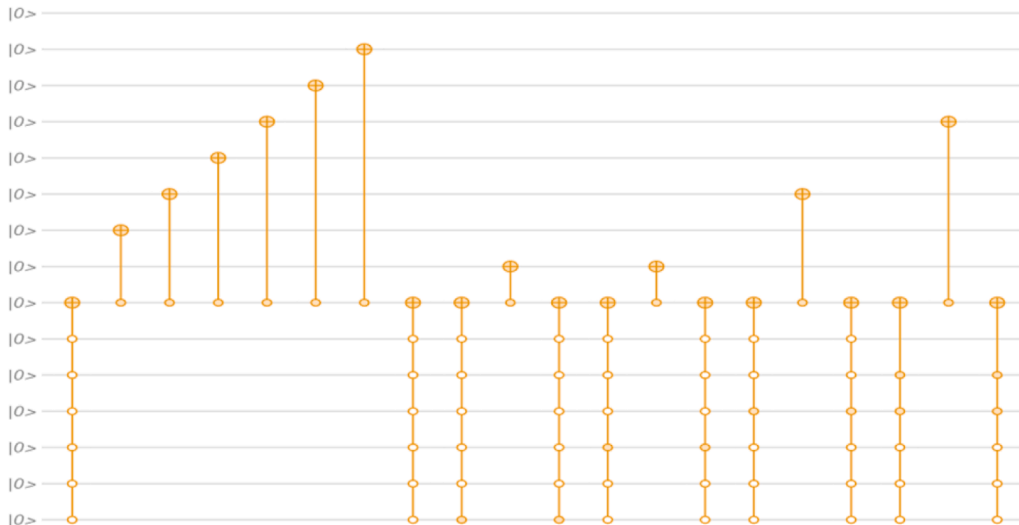


Fig. 9. Quantum circuit of DCT-EFRQI [8] for a 8×8 image, encoding DCT-transformed coefficients with ancilla qubits and Toffoli gates.

and position (Fig. 8).

$$|\psi_{\text{ENEQR}}\rangle = \frac{1}{2^d} \sum_{Y=0}^{2^d-1} \sum_{X=0}^{2^d-1} |P(Y, X)\rangle |0\rangle |YX\rangle \quad (40)$$

This model necessitates the use of $2d + k + 1$ qubits for storing grayscale images by using additional qubits to alter the position of pixels. It loads the position of additional qubits and sets grayscale values for the pixels using $k-2d$ and CNOT gates, respectively. The additional qubits are then reset to be used by another pixel, reducing time complexities.

- Nevertheless, FRQI inspires the DCT-EFRQI model [8], which uses quantum DCT and DWT to transform an image's pixel values into coefficients before encoding them as quantum states. The process has classical and quantum parts: the classical part transforms pixel coefficients, while the quantum part represents the quantized coefficients (Fig. 9). DCT-based compression reduces complexity, and ancilla qubits help link the coefficients to qubits using Toffoli gates. The method requires $k + 2d + 1$ qubits. The model involves applying and quantizing DCT, encoding the coefficients, and reconstructing the image. Double compression results in better efficiency than other models [8].

- Moreover, the IFRQI model [101] addresses the limitations of FRQI and NEQR by using two entangled qubit sequences. The first sequence uses l qubits to store $2l$ -bit pixel information, while the second functions like FRQI. The limitations include: 1) FRQI uses only one qubit per pixel with quantum superposition, and 2) NEQR uses $2l$ qubits for $2l$ -bit pixels without superposition. Although IFRQI has a lower time complexity than NEQR, it is slightly higher than FRQI. This model uses rotation matrices based on selected angles to retrieve image information through projective measurements. A $2^d \times 2^d$ quantum image with 2^{2l} gray range is formulated as follows:

$$|\psi\rangle = \frac{1}{2^d} \sum_{j=0}^{2^{2d}-1} \otimes_{q=0}^{q=l-1} (\alpha_{j,q}|0\rangle + \beta_{j,q}|1\rangle) \otimes |j\rangle \quad (41)$$

where, $\alpha_{j,q} = \cos\theta_q$, $\beta_{j,q} = \sin\theta_q \forall [0 \leq q \leq l-1]$.

2.11. FTQR

- Another approach for representing quantum digital images [105] highlights the inefficiency of traditional pixel-based methods for handling high-dimensional data. They suggest a Fourier transform-based method, which reduces data dimensionality and allows more efficient processing. The model applies a qubit-based Fourier transform for grayscale images and a three-point DFT for color images using superposition. A $D \times B$ grayscale image ($g_{d,b}$) in the 0 to 255 range is represented by the exponential coefficients, followed by its Fourier transform representation using $d \times b$ qubits:

$$g_{d,b} \rightarrow T[g_{d,b}] = e^{i\alpha g_{d,b}} \quad (42)$$

$$\bar{f} = \frac{1}{\sqrt{DB}} \sum_{d=0}^{D-1} \sum_{b=0}^{B-1} T[g_{d,b}] |db\rangle = \frac{1}{\sqrt{DB}} \sum_{d=0}^{D-1} \sum_{b=0}^{B-1} e^{i\alpha g_{d,b}} |db\rangle \quad (43)$$

However, for color images, a similar approach is used for each RGB color channel using $d \times b$ qubits as follows, wherein $\alpha = \frac{2\pi}{1024}$:

$$\begin{aligned} \bar{r} &= \frac{1}{\sqrt{DB}} \sum_{d=0}^{D-1} \sum_{b=0}^{B-1} e^{i\alpha r_{d,b}} |db\rangle, \bar{g} = \frac{1}{\sqrt{DB}} \sum_{d=0}^{D-1} \sum_{b=0}^{B-1} e^{i\alpha g_{d,b}} |db\rangle, \text{ and } \bar{b} \\ &= \frac{1}{\sqrt{DB}} \sum_{d=0}^{D-1} \sum_{b=0}^{B-1} e^{i\alpha b_{d,b}} |db\rangle \end{aligned} \quad (44)$$

2.12. QRMW

- The QRMW method [99] uses the principles of quantum mechanics

to encode the information of multi-wavelength and distinct-sized images into quantum states. All pixels' location, wavelength, and color scores are stored in a superpositional three-qubit sequence by employing I and H gates. The general representation of a $2^d \times 2^b$ image using this model is given as:

$$|\psi\rangle = \frac{1}{\sqrt{2^{q+d+b}}} \sum_{\lambda=0}^{2^q-1} \sum_{y=0}^{2^d-1} \sum_{x=0}^{2^b-1} |g(\lambda, y, x)\rangle \otimes |\lambda\rangle \otimes |yx\rangle \quad (45)$$

where, $|g(\lambda, y, x)\rangle = v_{\lambda y x}^0 \dots v_{\lambda y x}^{k-1}$, $v_{\lambda y x}^i \in \{0, 1\}$, $|g(\lambda, y, x)\rangle \in \{0, 2^k - 1\}$, vn denotes the count of wavelength channels for 2^k color range channel (2^k is the max value at any wavelength), and information of the channel and position is represented by λ and yx , respectively. However, it incorporates q ($q = \text{ceil}(\log_2 vn)$), k , & $d+b$ qubits that correlate to wavelength, color, and position information, respectively. The model has been demonstrated over an image with attributes such as size (2×2), color range (0 to 255), and several channels (4), using a total of 12 quantum computational bases wherein in the 2nd and 3rd sequence of qubits, the information of color and position is retained, respectively.

2.13. BRQI

- A quantum grayscale and a color image are constructed from 8 and 24-bitplanes, respectively, wherein the former can be decomposed to eight and the latter to 24 (or three-grayscale) binary images. This is accomplished by the BRQI model [97], which is formulated as:

$$|\psi\rangle = \frac{1}{\sqrt{2^{d+3}}} \sum_{u=0}^{2^3-1} \sum_{x=0}^{2^{d-b}-1} \sum_{y=0}^{2^b-1} |f(x, y)\rangle |x\rangle |y\rangle |u\rangle \quad (46)$$

However, the color image can be decomposed into three bitplanes as follows:

$$\begin{cases} |\psi_b^r\rangle = \frac{1}{\sqrt{2^{d+3}}} \sum_{u=0}^{2^3-1} \sum_{s=0}^{2^{d-b}-1} \sum_{t=0}^{2^b-1} |f_r(x, y)\rangle |x\rangle |y\rangle |u\rangle \\ |\psi_b^g\rangle = \frac{1}{\sqrt{2^{d+3}}} \sum_{u=0}^{2^3-1} \sum_{s=0}^{2^{d-b}-1} \sum_{t=0}^{2^b-1} |f_g(x, y)\rangle |x\rangle |y\rangle |u\rangle \\ |\psi_b^b\rangle = \frac{1}{\sqrt{2^{d+3}}} \sum_{u=0}^{2^3-1} \sum_{s=0}^{2^{d-b}-1} \sum_{t=0}^{2^b-1} |f_b(x, y)\rangle |x\rangle |y\rangle |u\rangle \end{cases} \quad (47)$$

where, $f(x, y)$, $f_r(x, y)$, $f_g(x, y)$, $f_b(x, y) \in \mathbb{C} = \{0, 1\}$, and the BRQI representation for color images is formulated as follows:

$$\psi_b^c = \frac{1}{\sqrt{3}} (|\psi_b^r\rangle |01\rangle + |\psi_b^g\rangle |10\rangle + |\psi_b^b\rangle |11\rangle) \quad (48)$$

Nevertheless, this model extends storage capacity by 16 and 2^{18} times for grayscale and color images, respectively [97]. The BRQI quantum circuit of a 2×2 image is represented in Fig. 10

2.14. NASS

- The NASS model [90] addresses challenges in representing and processing d -dimensional color images by using $d+1$ qubits wherein d represents the color and coordinates of 2^d pixels. It reduces the need for more qubits and improves image segmentation for content-driven searches. The $d+1$ and $d+2$ qubit states are illustrated for relative phases (NASSRP) and three components (NASTC), respectively, with the latter employed for lossy and lossless compression. The model, as formulated below, uses a bijective function for one-to-one mapping and defines color through an angle coefficient:

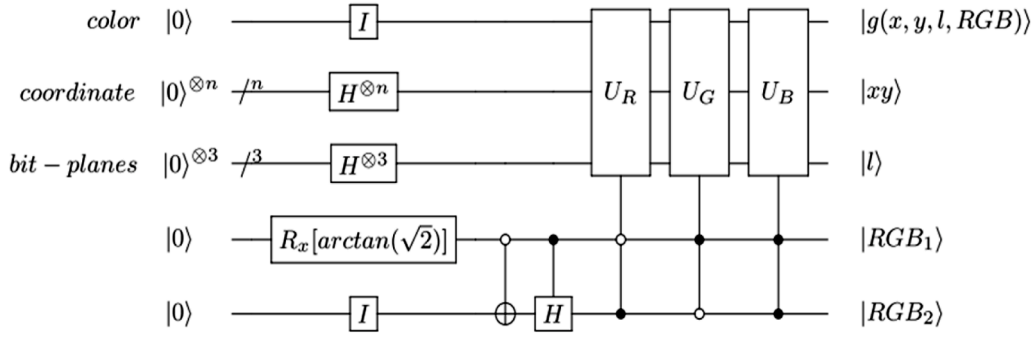


Fig. 10. BRQI quantum circuit [97] for a 2×2 color image. It requires $2n + 6$ qubits to store a $2^d \times 2^d$ color image.

$$|\psi_R\rangle = \sum_{k=0}^{2^d-1} \theta_k |v_1\rangle |v_2\rangle \dots |v_l\rangle |Y_k\rangle \quad (49)$$

where, $|Y_k\rangle = \cos\beta_k|0\rangle + e^{i\alpha_k}\sin\beta_k|1\rangle$ represents additional pixel information of $|k\rangle$ coordinates with $e^{i\alpha_k}$ relative phase of $|Y_k\rangle$. Furthermore, this model, when compresses a RGB image, is formulated as follows:

$$|\psi_C\rangle = \sum_{k=0}^{2^d-1} (\theta_{kR}|k\rangle|01\rangle + \theta_{kG}|k\rangle|10\rangle + \theta_{kB}|k\rangle|11\rangle) \quad (50)$$

2.15. OQIM

• Lastly, the OQIM approach [83] uses a sequence of quantum computational bases to hold the ascending order of every pixel using the magnitude of grayscale values. It uses the amplitude probability of two qubits to store color and position information, respectively. The model, over an image of size $2^d \times 2^d$, is formulated as:

$$|\psi\rangle = \frac{1}{2^{d+1/2}} \sum_{l=0}^{2^d-1} (\cos\theta_l|00\rangle + \sin\theta_l|10\rangle + \cos\theta_l|01\rangle + \sin\theta_l|11\rangle) \otimes |l\rangle \quad (51)$$

• In addition, this model is compatible with only single images; however, an enhanced variant mOQIM [83] is proposed to deal with multiple images as follows:

$$|\psi\rangle = \frac{1}{2^{d+1/2}} \sum_{l=0}^{2^d-1} (\cos\theta_{k,l}|00\rangle + \sin\theta_{k,l}|10\rangle + \cos\theta_{k,l}|01\rangle + \sin\theta_{k,l}|11\rangle) \otimes |l\rangle \quad (52)$$

Nonetheless, it necessitates $2d+2$ and $2d+q+2$ qubits with complexities $O(2^{4d})$ and $O(2^{4d+2q})$ for OQIM and mOQIM models, respectively.

2.16. Quantum image encoding

Beyond quantum image representation models, quantum encoding/

$$|l\rangle = \frac{1}{\sqrt{2}}|00\rangle|00000000\rangle + \frac{1}{\sqrt{2}}|01\rangle|01010110\rangle + \frac{1}{\sqrt{2}}|10\rangle|10101111\rangle + \frac{1}{\sqrt{2}}|11\rangle|11111111\rangle \quad (55)$$

embedding techniques map classical data, such as image pixels, into quantum states. They enable compact and high-dimensional representations while leveraging quantum parallelism for faster quantum computations. This makes them valuable for image classification tasks in quantum machine and deep learning [69,113,114], where they enhance

feature extraction and decision-making processes. By embedding classical image data into a quantum framework, these techniques integrate quantum computing with AI. In addition, they help optimize quantum circuits for practical implementations, reducing qubit overhead and computational complexity. Key quantum encoding methods include:

1) *Basis Encoding*: It maps pixel values to qubits using the amplitudes of quantum computational basis states ($|0\rangle$ and $|1\rangle$). It usually caters to binary images; however, grayscale images can be converted to binary vectors using the binary thresholding method. Each resulting vector is then mapped to a corresponding quantum state, and the superposition of these states represents the entire image. It is formulated as:

$$|\psi\rangle = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |p_i\rangle \quad (53)$$

where, p_i represents the binary pixel value, $|p_i\rangle$ represents its corresponding quantum state, and d is the total number of pixels in the image.

2) *Superposition Encoding*: Quantum superposition allows qubits to exist in multiple states simultaneously, enabling the encoding of pixel values as a superposition of states. It creates a quantum state as a linear combination of basis states. For a $d \times d$ image, it is formulated as:

$$|\psi\rangle = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |i\rangle |p_i\rangle \quad (54)$$

where, $|i\rangle$ is the pixel position and $|p_i\rangle$ encodes the pixel value. For a 2×2 grayscale image with four pixels (0, 86, 175, and 255, binary encoded as (00000000, 01010110, 10101111, and 11111111, respectively)), the superposition encoding can be represented as:

3) *Amplitude Encoding*: It maps a classical image into the amplitudes of a quantum state, leveraging superposition for efficient representation. Unlike basis encoding, which relies on measurement probabilities, amplitude encoding directly encodes pixel values into

quantum state amplitudes. For an image with d pixels, it requires $\log_2 d$ qubits to encode all pixels. Mathematically, it is formulated as:

$$|\psi\rangle = \sum_{i=0}^{d-1} \alpha_i |i\rangle \quad (56)$$

where, α_i is the complex amplitude associated with each pixel, $|i\rangle$ represents the basis state of each pixel, and the amplitudes satisfy the normalization condition. While it efficiently compresses data into a quantum state, its implementation is computationally expensive. The required circuit depth scales as $O(2^d)$ for d qubits, making it challenging for large datasets.

4) Angle Encoding: It maps pixel values to phase angles in quantum states, where phase shifts correspond to rotations in the Hilbert space. These rotations are applied through quantum gates around the X, Y, and Z axes to alter the quantum state. For an image pixel p , the quantum state is represented as:

$$|\psi_p\rangle = R_y(\theta(p))|0\rangle \quad (57)$$

where, $R_y(\theta(p))$ is the rotation operator around the Y-axis acting on the $|0\rangle$ state, and $\theta(p)$ is the angle determined by the pixel. Angle encoding reduces circuit depth to $O(1)$ with one gate per qubit. However, it requires $O(d)$ qubits for an image with d pixels, leading to $O(2^d)$ amplitudes. The rotation operators are defined as:

$$R_x(\theta(p)) = \begin{bmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{bmatrix}, R_y(\theta(p)) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}, \text{ and } R_z(\theta(p)) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix} \quad (58)$$

5) Dense Angle Encoding: It encodes two pixels per qubit to solve the issues in amplitude and angle encoding methods [69]. It reduces the qubit requirement to $d/2$ and the amplitudes to $O(2^{d/2})$. The second pixel in each qubit is encoded with a phase shift gate P .

$$|\psi_x\rangle = R_y(\theta_1)P(\theta_2)|0\rangle \quad (59)$$

6) Hybrid Encoding: It combines amplitude and angle encoding techniques to balance qubit usage and circuit complexity, improving model efficiency [41,115].

7) Ground State Encoding: It represents image pixels by associating them with a quantum system's lowest energy state (ground state). The binary pixel values are mapped directly to the computational basis states, i.e., $0 \rightarrow |0\rangle$ and $1 \rightarrow |1\rangle$. It uses the system's intrinsic properties rather than relying on rotations or amplitudes.

8) Hamiltonian Encoding: It represents image pixels as quantum states by mapping their intensity values to the eigenvalues of a

Table 3
Types of quantum noise [16].

Noise Type	Effect
Bit flip	Variation in the computational state from $ 1\rangle$ to $ 0\rangle$ or vice versa with some probability upon measurement introduces bit flip noise in the system.
Phase flip	Phase error causes rotation of qubit around the Z axis, which results in $(\alpha 0\rangle + \beta 1\rangle) \leftrightarrow (\alpha 0\rangle - \beta 1\rangle)$. It is like the phase-damping channel [118] and is the only coherent error.
Depolarizing	It causes loss in phase, and the superposition of basis states results in decay into mixed states. It causes sudden death of maximally entangled qubits and reduces concurrence to 0.
Amplitude damping	The decay of qubits from an excited state $ 1\rangle$ to a ground state $ 0\rangle$ irreversibly causes amplitude-damping noise in the quantum system.
Pauli Channels	Pauli Channel noise occurs when Pauli operators are randomly applied to quantum channels, which may cause entanglement and sudden death.
Two-qubit depolarizing	It causes a loss of superposition between the 2-qubit basis state and its quantum phase due to depolarization by spontaneous transition among eigenstates.
Two-qubit dephasing	It causes a loss of superposition between a 2-qubit basis state and its quantum phase due to dephasing (i.e., loss of phase coherence).

Hamiltonian, H , which governs the quantum system's time evolution. For an image with pixel intensities, p_i , the H is defined as:

$$H = \sum_{i=0}^{d-1} p_i |i\rangle \langle i| \quad (60)$$

where, $|i\rangle$ denotes the quantum state corresponding to the pixel position i and p_i defines the eigenvalues of H . It is used for matrix inversion in algorithms like PHLA [42].

However, the general workflow of a quantum machine learning process using any of the above encoding techniques is outlined in Fig. 11 below.

These quantum image representation models have driven researchers to explore their applications in various image processing tasks, evaluating their feasibility, computational efficiency, and potential advantages over classical methods. This exploration aims to leverage quantum properties for novel capabilities that surpass the limitations of traditional approaches.

3. Quantum image processing algorithms

This section presents quantum image processing algorithms that offer faster and more accurate image analysis. These connect quantum computing to real-world applications like denoising, edge detection, cryptography, and classification, offering innovative solutions to



Fig. 11. Workflow of quantum machine learning.

complex image processing challenges.

3.1. Quantum image denoising

Noise represents a fundamental challenge in building large-scale quantum computers and algorithms [116]. It significantly hinders the successful implementation of quantum image processing methods [117].

- 1) In the physical realm, noise arises due to unwanted interactions between the qubits and the environment, imperfect operations, and leakage [118].
- 2) Images are susceptible to different noise types, accumulating at multiple vulnerable points, including the sensors, digitization processes, and potential interference during transmission [118]. It refers to unwanted random variations in pixel values from images represented, processed, and stored in a quantum computing framework. Images can be influenced by environmental factors such as low light conditions, fog, and temperature variations. However, image noise can take various forms depending on the effects and patterns, like Gaussian, Salt and Pepper (Impulse), Speckle, Poisson, [119] etc.
- 3) The induction of error in the quantum system can be classified as coherent and incoherent [118]. Coherent error introduces unitary operation's alteration to the input pure state; though the purity of the input state is preserved, this error keeps adding to the quantum circuit. An incoherent error is caused by interaction with the environment and changes the quantum system's state to a mixed state. Various types of quantum noise [120] and their effect are described in Table 3.
- 4) Nonetheless, noise can be of two types [119], depending upon the source: *Additive*: $M(d) = I(d) + N(d)$ and *Multiplicative*: $M(d) = I(d)N(d)$, where, $I(d)$ denotes an image with no noise, $N(d)$ denotes the noise, and $M[d]$ represents the noisy image.

However, filtering techniques are employed to mitigate the impact of noise that smoothen pixel values by averaging neighboring pixels. This smoothing process effectively reverses the noise-induced distortions. Denoising, a.k.a. noise reduction, one of its kind, enhances the visual quality of images, making them more suitable for various applications in quantum image processing. The prime concern of denoising is to recover $I(d)$ from $M(d)$ with a higher fidelity score, tending to one. In essence, denoising returns an image to its original, pristine state, safeguarding all its vital information. Additionally, denoising can be achieved in two distinct filtering variants [119]:

- 1) *Linear denoising*: It offers speed and requires less computational time but may entail a trade-off in preserving fine image details and structure. It is evident when considering techniques like mean (with sliding or averaging window), nonlocal mean, and anisotropic mean filters.
- 2) *Non-linear denoising*: In contrast, it is exceptionally proficient at suppressing noise and maintaining image quality with sophisticated and recursive noise reduction operations, albeit at the expense of increased processing time. The examples include median (with varying window size as per noise) and bilateral filters, which preserve smooth regions or texture details [121].

In the realm of classical approaches, extensive research has been conducted; however, in quantum computing, the journey has been initiated and accelerated with the advent of the MFIQC approach [122]. It introduces a quantum-inspired superposition of structuring elements for morphological image processing. It is a linear superposition of all structuring elements within a size range and collapses into a single element upon measurement. New collapsing morphological operators are defined based on this concept, leading to a self-adaptive filtering method. This method uses a 3×3 structural element with a variable

mask that adapts to its location in the image. Simulations show that this filter is more effective in denoising than median and traditional morphological filters. It performs well for low-intensity noise but struggles when noise intensity increases, leading to significant performance deterioration [123].

To address this issue, the QCMF method [123] extends MFIQC for varying noise intensities. It presents a quantum collapsing median filter for Salt and Pepper noise reduction. It collapses possible and non-possible noisy pixels to states 0 and 1, respectively. It adaptively adjusts the window size from 1×1 to 11×11 based on noise intensity. Finally, Pixels with superposition state 1 in this window undergo median filtering. Its results were validated over an image with 90 % of the noise, resulting in a higher PSNR than classical standard and adaptive median filters [123].

The practical significance of denoising becomes evident in QNCDT [124], where a quantum wavelet denoising algorithm is applied to enhance chest digital tomosynthesis. It uses decomposition to divide an image into four frequency components: smoothing-LL, horizontal-HL, vertical-LH, and diagonal-HH (L and H indicate low and high frequency components). It transforms a wavelet into an image at each level by repeating the division for a smoothing image. Using Daubechies' mother wavelet function, a wavelet-transformed image is constructed, followed by balance sparsity norm and thresholding for effective noise reduction.

X-ray fluoroscopic images were also tested to improve their quality by employing various denoising algorithms [125] to mitigate Poisson noise. Among these, BM3D proved effective, using collaborative filtering through sparse representation and transform domain. The method groups similar image regions based on sparsity, applies a 3-D transformation to 2-D image blocks, shrinks the transform spectrum, and performs an inverse 3-D transformation. It produces a 3-D estimate of jointly filtered grouped image blocks. Comparative results highlight collaborative denoising as the most effective strategy.

Median filtering is a common choice for image processing, but its denoising hinges on filter mask size. A small mask preserves edges but offers weak denoising, while a large one excels at denoising but blurs details [117]. Striking the right balance, especially in noisy images, is challenging. An adaptive median filter, QAMF [117], is presented to address this issue. It transforms pixels into a dual qubit state, enriching their representation with four basis vectors to improve precision in pixel manipulation and suppress the Impulse noise. However, to boost the efficiency of median filters, the QMFSD model [126] performs $2^d \times 2^d$ median calculations over 9 images simultaneously.

The QFSD model [127] introduces a quantum spatial domain image filtering method that uses quantum addition for sequence correlation instead of multiplication which cannot be realized physically. It includes quantum circuits for shifting positions and a sequence of quantum image addition and subtraction operations. It filters grayscale images and effectively correlates with the filter mask. The time complexity of the quantum correlation ($O(k^2(d^2 + q^2))$) is lower than its classical variant ($O(2^{2d})$). Further, a Median filter model was also tested for Gaussian, Impulse, and qubit flip noise [126]. Nevertheless, these approaches suffer from computational slowness, prompting the development of an alternative model, QFFD [128]. It uses the quantum Fourier transform to process an image, applies a filter function via quantum oracle, and applies its inverse to recover the original image. QFFD achieves significantly faster computation compared to the classical fast Fourier transform, offering an efficient approach to image denoising.

Furthermore, the QMLSONN model [129], analogous to classical MLSON, works with binary images. It consists of three layers, input, hidden, and output, where qubit-based neurons replace classical nodes. However, qubit rotation gates act as weights for the interconnections between layers. The aggregated qubit information from multiple nodes is processed through a neighborhood topology-based interconnection matrix, which employs a characteristic sigmoidal activation function to handle the nonlinearities in system inputs. After measuring the output

Table 4

Comparison of quantum image denoising techniques, wherein only one model operates on multiple images simultaneously, while only two models operate on color images.

Model [Ref.]	Selection Mask	Type of Noise	Type of Filter	Image Type	N*
QCMF [123]	$(1 \times 1 - 11 \times 11)$	Impulse	Collapsing Median	N/A	1
QAMF [117]	N/A	Impulse	Adaptive Median	N/A	
QBID [130]	Horizontal Rafter (3×3)	Classical	Boolean Mean		
QCLFE [121]	Convolution Mask (3×3)	Gaussian	QC-LFE		
DQWT [20]	N/A	Gaussian and Impulse	Quantum Wavelet		
QFSD [127]	Neighborhood (3×3)	Impulse	Quantum Correlation		
QFFD [128]	(3×3)	N/A	Quantum Fourier and its Inverse		
QMFSF [126]	(3×3)	Gaussian, Impulse, and qubit flip	Median		
QMLSONN [129]	N/A	Gaussian and Uniform	Backpropagation		
QNCDT [124]	N/A	N/A	Balance sparsity-norm, thresholding		1
QMBT [131]	$5 \times 5, 7 \times 7$ and 11×11	Gaussian and Poisson	Thresholding, backpropagation		
XRFI [125]	N/A	Poisson	AAS, AV, BM3Dc		

* It represents the number of images considered for denoising simultaneously.

states to identify errors, quantum backpropagation is used to optimize performance. Results show that this model outperforms classical models like MLSONN and the supervised Hopfield network [129]. Thereby, highlighting the potential of quantum machine learning for advanced noise reduction in quantum images.

However, the mean filter, implemented in internal representation for quantum images, is put forth by the QBID model [130], which entails the following steps: 1) decompose colors by separating the RGB channels of a noisy image into bit planes, 2) use the MSB plane from each channel for processing between classical and quantum representations, 3) apply the QBMF method [130] to denoise the MSBs of each color channel, 4) reconstruct the image using the denoised MSBs from each channel, and 5) combine the denoised MSBs with the untouched bit planes to reassemble the final image. Comparative analysis reveals that the QBID model surpasses the classical mean filter in performance metrics such as MAE, MSE, and PSNR, demonstrating its superior noise reduction capabilities.

To address Gaussian noise with different noise levels ($\sigma = 15, 20, 25, 30$), the QCLFE model [121] emphasizes the utilization of quantum calculus of local fractional entropy, wherein the former involves assessing the probability of each pixel. It entails: 1) initialize 3×3 masks, (2) set fractional parameter values, (3) add Gaussian noise for evaluation, (4) convolve corrupted image pixels with the mask, (5) apply the proposed and Gaussian filters for noise removal, (6) assess effectiveness using PSNR, and (7) extend the method to color images by processing RGB channels separately. However, the experimental outcomes show that the highest PSNR value was achieved over Lena's image, with $\sigma = 15$, surpassing the standard Gaussian filter.

Unlike the previous methods, the DQWT model [20] embeds a noisy image into the wavelet coefficients of the original image. It extracts these coefficients from a 4th-order quantum Daubechis kernel [20]. Subsequently, a quantum oracle is applied as a thresholding function to

decompose the wavelet coefficients. It enhances the original image while minimizing the impact of noise on its wavelet coefficients. Results show that DQWT outperforms both its classical counterpart and other quantum-based denoising approaches while requiring fewer qubits, making it more efficient in quantum resource utilization.

Nevertheless, the quantum many-body denoising algorithm, introduced by QMBT [131], leverages the concept of interaction from many-body physics to tackle denoising challenges in image processing. This algorithm selects a target patch within a search window and extracts all neighboring patches. Subsequently, it calculates their interaction and effective potential (combining the target patch with total interaction). It enables the generation of adaptive vectors through many-patch interactions that adhere to Schrödinger's equation, offering patch-based denoising and overall image enhancement. Finally, the effectiveness of this approach is validated through medical ultrasound image despeckling, showcasing its real-world application. Nevertheless, the comparison between all the denoising techniques is presented in Table 4.

3.2. Quantum edge detection

Edge detection is the foundation of numerous computer vision tasks, from pattern recognition to image segmentation and OCR detection. It offers significant advantages in medical imaging, biometric authentication (including face and fingerprint recognition), historical document restoration, autonomous vision systems, satellite imagery analysis, and more. It relies on identifying edges, which trace the boundaries within an image, marking abrupt shifts in illumination or variations in grey-level magnitudes [119]. These variations encompass depth, illumination, surface normals, and color changes. Edge detection operates on the premise that these discontinuities can be quantified by computing the first derivative of an equation that describes intensity variations in an

image. Moreover, the gradient highlights the presence of discontinuities and provides valuable insights into their direction and magnitude, which is crucial for subsequent analysis and interpretation [119].

In traditional settings, edge extraction finds its footing in algorithms such as Prewitt, Canny, Sobel, Robert, Laplacian of Gaussian, Kirsch, and more [119]. Over time, researchers have ventured into quantum variants, introducing innovative approaches to edge detection. One notable adaptation is by [132], a quantum rendition of the Prewitt algorithm. Using averaging differences, the Prewitt mitigates edge blur while preserving a degree of smoothing and sharpness along the edge's direction. It entails vertical (P_v) and horizontal (P_h) operators to identify edges. In a significant advancement, Duby et al. [132] have translated Prewitt's method into a quantum framework, yielding notable improvements in edge continuity. Each pixel is transformed into a qubit with state probabilities calculated using Prewitt operators to determine the gradient components. The magnitude of the gradient vector is then computed, and the probability of edge presence is characterized by P_v and P_h , ensuring that higher gradient values correspond to greater

primarily targeting the most significant bit of an image. The approach relies on a convolutional mask applied in a horizontal raster manner, interacting with a portion of the image to enhance edges. The classical version follows three main steps: 1) applying an OR operation on all mask elements except the center, 2) performing an XOR operation between the result and the center pixel, and 3) replacing the center pixel with the computed value. The quantum version extends this method using the PAO criterion, leveraging quantum logic in place of Boolean operations to achieve similar edge detection but with potential quantum speedup.

The classical Sobel method uses discrete differentiation to estimate the gradient of an image. It applies 3×3 vertical (S_v) and horizontal (S_h) masks as first-order derivatives to detect edges in their respective directions. It accurately detects and highlights edges in the image by calculating pixel magnitudes and identifying the strongest transition from light to dark [134].

$$S_h = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, S_v = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}, P_h = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}, P_v = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix} \quad (61)$$

probabilities. In the mapping output stage, if the state is $|0\rangle$, the output is 0, while $|1\rangle$ allows for either 0 or 1 based on specific conditions. The final result is a bi-level edge image after processing all pixels. Evaluation over Lena's image demonstrated the supremacy of this model over various classical models like Sobel, Canny, LoG, and Robert.

Nevertheless, Mastriani proposed a model [133] with its initial step as a pole-to-pole axis only that constitutes projections on the vertical axis of the Bloch sphere. It presents a novel edge detection method for both classical Boolean and quantum Boolean image processing,

Table 5

Comparison of quantum edge detection methods, wherein no model operates on color images. However, QSobel, MarrHED, and QRED, have lowest complexity with MIHE having the highest complexity.

Model [Ref.]	Extraction method	Image Type	Representation	Complexity
EEKO [138]	Kirsch		NEQR	$O(d^2 + 2^{k+3})$
BQTHA [135]	Hilditch			N/A
QPrewitt [132]	Prewitt	N/A	N/A	$O(2^{2d})$
LoG [117]	Gaussian-Laplacian		N/A	N/A
QBED [133]	Boolean		Boolean	N/A
QSobel [134]	QSobel		NEQR	$O(d^2)$
MarrHED [136]	Marr Hildreth			$O(d^2)$
MIHE [140]	Hill Entropy			$O(2^{4d})$
LZCM [137]	Laplacian			N/A
QRED [139]	Robinson			$O(d^2 + 2^{k+3})$

The quantum adaptation of classical Sobel's model is presented as QSobel [134]. It works as: 1) transform a classical image into an FRQI quantum image using a $(2d + 1)$ -qubit register, with additional qubits allocated for neighborhood pixel color information and results. 2) Apply X-Shift and Y-Shift transformations to extract the color qubits of neighboring pixels and store them in an extra qubit. 3) Utilize the quantum black box to compute Sobel gradients for all pixels simultaneously, storing the results in another qubit. The final quantum image distinguishes edges, where pixels in the edges have a color qubit $|1\rangle$ (white), while non-edge pixels have $|0\rangle$ (black), effectively extracting edges from the quantum image. When compared to the existing quantum and classical algorithms, the QSobel surpasses them all.

Skeletonization (a.k.a. thinning) is a process applied to binary images aimed at preserving only the connectivity of the image by selectively discarding pixels [135]. There are various such algorithms in a classical setting, while the first in quantum was proposed by BQTHA [135], a quantum variant of the classical Hilditch's algorithm. It operates on specific criteria: identified pixels must qualify as edge points, not endpoints or breakpoints, and the resulting pattern should exhibit optimal thinness while maintaining centrality and connectivity.

In the LoG method [117], convolution with a Gaussian operator is applied to smoothing pixels, effectively reducing noise and creating a more uniform intensity distribution. Subsequently, a Laplacian operator is utilized to discern variations in grey-level magnitudes across pixels. It identifies regions of rapid intensity change, highlighting potential edges within the image. Notably, pixels where the Laplacian operator yields zero crossings are identified as edges, indicating significant transitions in intensity and facilitating precise edge detection [117]. Similarly, the classical Marr-Hildreth edge detection method is transformed into the quantum domain by [136], which entails Gaussian-Laplacian filtering and zero crossing extraction. Another model [137] uses a Laplacian filter and zero-crossing extraction based on similar foundations.

Further, a quantum rendition of the Kirsch algorithm is presented by [138]. Operating across eight directions, it identifies the longest directional response as the edge within the image in four steps: 1) acquire the classical image, 2) transform 1) to quantum setting using NEQR [19], 3) the gradient of 2) is calculated by the Kirsch matrix, and 4) the edges in

3) are extracted using the threshold operation [138].

$$|t\rangle = \frac{1}{2^d} \sum_{k=0}^{2^d-1} |Tyx\rangle |yx\rangle \quad (62)$$

where, $Tyx \in \{0, 1\}$; however, when the output is 1, the edge can be extracted [138].

Similarly, based on the classical Robinson model, [139] employs a mask rotated in all eight directions based on the 0th column. As this approach increases the accuracy of the gradient, it requires more computation than the Sobel operator [139].

Furthermore, the researchers [140] present an approach for edge detection in medical images, addressing the challenges of dark backgrounds and low contrast. It introduces an enhanced quantum representation model that improves grayscale values using Caraiman's. Quantum noise is tackled using a neural network-based hybrid filter, eliminating noise effectively. Additionally, a novel edge extraction based on Hill entropy reduces computation time while maintaining high-quality results. The edge detection is accomplished in the following steps: 1) evaluate histogram and normalize to get a probability distribution, 2) $\forall$ threshold (k): divide the image into $R1$ ($\{x|x \leq k\}$) and $R2$ ($\{x|x > k\}$), 3) formulate the Hill entropy of each region:

$$H = \frac{1}{1-\alpha} \log \left(\sum_{x \in X} p(x)^a \right) \quad (63)$$

4) calculate the weighted sum of entropies of $R1$ and $R2$, and 5) find the value of k that maximizes 4). This method separates the background and object. Nevertheless, various quantum edge extraction techniques are compared as follows in Table 5.

3.3. Quantum image transformations

Image transformations play a crucial role in computer vision, image analysis, and the strengthening of quantum image encryption schemes. A notable example is motion correction, which involves image registration and resampling using parameters derived from a reference image. Various image transformations, such as scaling, translation, and geometric adjustments, are extensively studied on classical computers [141]. However, on quantum computers, they remain relatively unexplored.

- 1) *Image scaling* employs interpolation to estimate values at unknown locations using known values. Its quantum variant was kick-started by [91] to improve NEQR [19], allowing the storage of images of size $2^d \times 2^b$ rather than $2^d \times 2^d$. Additionally, they extended nearest neighbor interpolation to the quantum domain, producing lower quality results than bilinear and bicubic methods, presenting a foundational step towards quantum image scaling research.
- 2) *Image translation* can be an entire translation moving the whole image or cyclic translation used in intermediate steps like shifting the zero-frequency component to the center of an image's spectrum post-Fourier transformation. Neither type of image translation had been extended to quantum computers using quantum adders and comparators until [142]. Based on NEQR [19], the algorithm applies to

$2^d \times 2^d$ grayscale images and processes left and right translations differently, increasing the network circuit complexity.

- 3) *Morphological transformations* are well explored in classical settings, wherein dilation and erosion are most common which are used in edge detection, gradient calculation, etc. However, research in the quantum domain remains relatively limited. Yuan et al. [143] presented two quantum dilation and erosion methods based on shifting positions and storing neighborhood pixel information within a quantum image set. Their approach demonstrated superior performance over classical methods, achieving higher efficiency with reduced computational complexity. Similarly, Li et al. [144] explored these operations on binary and grayscale images, proposing three quantum circuits designed for noise removal, boundary extraction, and skeletonization. The findings highlighted the potential of quantum morphological processing in enhancing image quality and structural analysis. Further, Fan et al. [145] proposed flat dilation and erosion operations for NEQR-based images showcasing efficient results.
- 4) *Geometric transformations* are crucial for encryption and image registration, involving resizing, rotating, and cropping operations. Relying on FRQI [18], various algorithms manipulate pixel positions through geometric transformations. One such approach, the GTQI [108] method, achieves this by swapping coordinates, flipping and rotating orthogonally, etc. It is constructed as follows:

$$g_j |I_{FRQI}(d)\rangle = \frac{1}{2^d} \sum_{j=0}^{2^d-1} (|k_j\rangle \otimes g|J\rangle) \quad (64)$$

where, I_{FRQI} denotes the FRQI image, $g|J\rangle$ denotes the unitary geometric operation being performed on $|J\rangle$, $|k_j\rangle$ and $|J\rangle$ denote color and position information, $j = 0, 1, 2, \dots, 2^d - 1$. In addition, rGTQI, or restricted GTQI [109,110], is intended to work with more complex transformations to construct robust yet advanced applications. The GTQI and rGTQI rely on manipulating spatial context, while other models like CTQI [111] focus more on manipulating the color path of FRQI [18]. The following is the formulation of the color transformation of each pixel in an image being controlled:

$$C_J |I_{FRQI}(d)\rangle = \frac{1}{2^d} \sum_{k=0}^{2^d-1} C(|c(\theta_k)\rangle) \otimes |i\rangle \quad (65)$$

where, C_J represents the color transformation qubit gate of $|I_{FRQI}(d)\rangle$ quantum image, $|c(\theta_k)\rangle = \cos(\theta_k)|0\rangle + \sin(\theta_k)|1\rangle$ is used for encoding color information.

Further, Le et al. [108] extended geometric operations like two-point swapping, flipping, coordinate swapping, and orthogonal rotations using basic quantum gates, such as NOT, CNOT, and Toffoli. Interestingly, unlike classical geometric transformations, where global operations are generally slower than local ones, the opposite holds true in their quantum counterparts.

Quantum image transformations can be utilized to manipulate image pixels in such a way that the original image becomes unrecognizable to unauthorized viewers. By using quantum scrambling, pixel

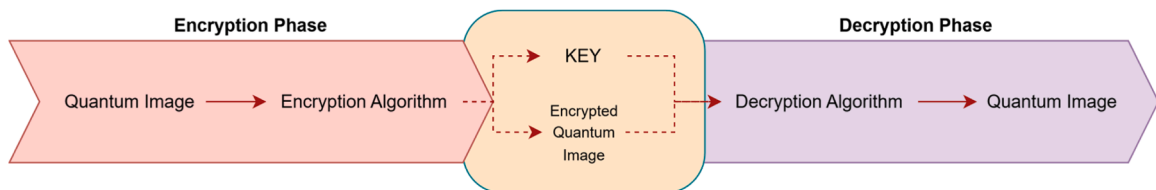


Fig. 12. The encryption phase converts a quantum image into an encrypted form using a quantum encryption algorithm. A secret key is generated and used during encryption, which is then transmitted. The decryption phase applies a quantum decryption algorithm using the same key to reconstruct the original quantum image.

transformations, and quantum encryption schemes, these models enhance image confidentiality, ensuring secure transmission and storage in applications like cryptography, steganography, and watermarking.

3.4. Quantum image security

Hiding and securing images or visual media while computing and communicating is crucial for preventing unauthorized access, modification, or distribution. Quantum image security incorporates quantum cryptography, steganography, and watermarking techniques to ensure confidentiality, integrity, and availability. These techniques leverage the principles of quantum mechanics, such as no-cloning, entanglement, and QKD, to provide unbreakable encryption. These are used to prevent quantum images from 1) unauthorized viewing or tampering, 2) unauthorized use or distribution, 3) improper control of access, and 4) interception or data breaches [119].

- 1) **Cryptography:** Encoding images to safeguard them during transmission.
- 2) **Steganography:** Concealing messages within public data.
- 3) **Watermarking:** Embedding ownership information in images to identify the rightful owner.

Cryptography encrypts the image content, whereas information hiding conceals its presence, which entails steganography and watermarking. While the latter two are interrelated, their purposes and applications differ. The carrier image contains visible content in watermarking, while copyright or ownership data is hidden for authentication. In contrast, steganography aims to discreetly convey secret messages within the carrier image. Watermarking typically embeds serial numbers or logos to denote ownership, whereas steganography often needs more space within the carrier image to conceal messages effectively. Additionally, watermarking is vulnerable to infringement risks like cropping or filtering, unlike steganography [71]. Quantum computers pose significant challenges to security, particularly for encryption methods like integer factorization based and elliptic curve-based cryptography [146]. However, leveraging principles of quantum physics can help mitigate these risks. Encryption protocols can exploit quantum uncertainty to provide intrusion detection and security against quantum computer attacks [147]. Furthermore, QKD offers promising solutions to safeguard sensitive information [148].

3.4.1. Cryptography

Cryptography involves encrypting image data with a key, as is depicted in Fig. 12, for secure transmission, ensuring only authorized parties can access it. Symmetric cryptography uses the same key, while asymmetric employs public and private keys to encrypt and decrypt data. However, the latter is deemed more secure due to the former's vulnerability to simple conversion operations that can result in obtaining the key [119]. Encryption is crucial for securing military communication and sensitive data while retaining fast computing speed, high fidelity, and large processing capacity for real-time video conferencing, precise disease diagnosis, and handling image data's strong correlation and redundancy. In quantum image encryption, the spatial domain is characterized by pixel position and color value; thus, algorithms primarily target these parameters, commonly employing scrambling to disorder pixel positions and replacement to alter color values [71].

- 1) **Scrambling:** Prominent scrambling algorithms utilized for position information include the Hilbert, Baker map [25], Arnold [98], and Fibonacci transforms [119], adapted for encryption in the quantum domain. The scrambling quantum circuits of the latter two leverage plain and adder module N to modify input and output operations, while the former employs a Hilbert scanning matrix generated via a recursive algorithm. In a preceding study [149], an encryption and decryption algorithm utilizing geometric transformations on FRQI

images aimed to reverse correlations among adjacent pixels; however, the resulting encrypted images lacked noise-like properties and could be visually identified. Furthermore, another algorithm [26] encodes classical images into quantum information using generalized quantum image representation and encrypts pixel values with keys generated from a four-dimensional chaotic system. The Arnold transformation is used to randomize pixel positions, and decryption reverses these steps. The algorithm produces scrambled images resilient to destructive attacks, overcoming issues like periodicity, limited key space, and susceptibility to statistical analysis [26].

- 2) **Replacement:** Besides scrambling, algorithms that incorporate or merge the replacement approach have been extensively used. Song et al. presented an encryption scheme [150] employing geometric transformations to shuffle pixel positions and recode color information in FRQI [18] images. Zhou et al. [151] introduced a quantum image Gray-code and bit-plane scrambling method based on NEQR representations, highlighting its cost-effectiveness and significantly faster than other techniques. They noted that while the circuit cost for the first category rises with image size, their scheme restricts the scrambled image to $d \times d$. Targeting MCQI-based RGB images, Yan et al. [152] apply color and geometric transformations, ensuring image security through quantum rotation operations for color information and an enhanced partition scrambling method for geometric data. Further, a novel quantum image encryption algorithm [153] was explored, combining the generalized affine transform with a logistic map. This algorithm encrypts gray-level information through XOR operations controlled by a logistic map-driven key generator, while position information is encoded using the generalized affine transform. This approach is deemed robust, with superior computational performance compared to classical methods.

An extended quantum variant of [154], introduced by [155], operates on a quantum Wavelet transform that unfolds in four steps: 1) diffusion is applied to the original image, 2) QWT is used on the quantum image, 3) another diffusion is used on the encrypted image, and 4) the inverse QWT is applied. The second QWT generates a chaotic logistic sequence via a general chaotic logistic map, ensuring uniform distribution of color information within the image. These diffusion operations involve unitary matrices, facilitating decryption by reversing the process.

Nevertheless, a novel method based on correlation decomposition correlates image pixels using quantum superposition and measurement, dividing the image into sub-images structured in a complete binary tree [156]. Color information is encoded within each sub-image using quantum random-phase and revolving operations, along with the Hadamard transform. The encrypted image is generated by superimposing these sub-images. The authors concluded that the algorithm is resilient against brute-force attacks, surpassing its classical counterpart in efficiency [156].

Building upon the concept of double-phase random encoding initially proposed by [157], Yang et al. [23] introduced its quantum variant. It transforms images into complex-valued encoded forms by applying random-phase encoding. However, Yang et al. [23] further strengthened security by implementing phase encoding on grayscale images across spatial and frequency domains. Notably, all operations in this encryption process are unitary, simplifying decryption through reversible processes. Moreover, its computational complexity is lower than its classical counterpart.

A quantum image encryption scheme employing geometric transformations by Zhou et al. [149] was found to violate quantum principles by Yang et al. [158], who then presented a colored image encryption algorithm using QFT with DPPE, offering improved security and computational efficiency. They also showcased the robustness of their algorithm through key sensitivity analysis, highlighting the practical challenge of obtaining the correct key due to its ample key space.

Further, the researchers [150] employed restricted geometric

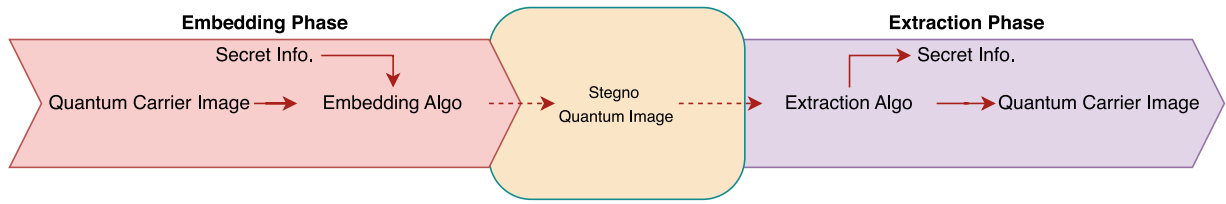


Fig. 13. In the embedding phase, a quantum carrier image undergoes processing through a quantum embedding algorithm, where secret information is embedded to form a steganographic quantum image. In the extraction phase, a quantum extraction algorithm retrieves the secret information while restoring the original quantum carrier image.

transforms to shuffle pixel positions and implemented a quantum diffusion operation based on restricted color transformations. Encryption keys were generated using sensitive chaotic maps, further enhancing security. However, other efforts in quantum image encryption involve diverse methods, such as a scheme leveraging a quantum logistic map, which was found to possess satisfactory speed, a sizable key space, and high security [159].

Furthermore, as per Dai et al. [160], QDCT is applied to four quantum grayscale images to generate coefficient matrices, which are compressed to form a new quantum image. A quantum key image is created using a 1D chaotic sequence from a 4D hyper-chaotic Henon map, XORed with the new quantum image. It is followed by pixel scrambling with logistic map-based cycle shifts, offering lower computational complexity than classical image encryption algorithms.

Liu et al. [26] utilize a 4D chaotic system to encrypt pixel values, followed by the Arnold transform to randomize pixel positions. For decryption, the process is reversed to recover the original image.

Similarly, Dai et al. [161] improved traditional pseudo-random number generators for encryption using a combination of quantum methods over FRQI-based quantum images. Initially, the neural network is optimized using the QPSO and trained over the samples based on quantum logistic map-based PRNG. The new sequences generated are involved in encoding phase angles using the quantum fractional Fourier transform followed by the scrambling operation. The inverse of QFFT and XOR operations are used to decrypt the image.

Hyper-chaotic systems have features that, when changed even by a small amount, can lead to significant changes in the results. The quantum cellular neural network, one of its kind, incorporates chaotic properties that affect other operations. Only a few qubits can lead to complex high-dimensional chaos that provides a unique level of security. In this context, the QCeNN [21] generates five hyper-chaotic sequences used in different operations to secure an image. Signal 1 controls the cycle shift of the image's gray pixels (shifting their positions). Signal 2 flips the gray pixels (changing their values). Signal 3 defines how many times the image undergoes an affine transformation. Finally, signals 4 and 5 control the translation coefficients. After all these operations, the result is a cipher image. The chaotic signals are converted back into integers to decrypt this image. The inverse affine transformation is applied to get the intermediate cipher image. The cycle shift, SWAP, and CNOT operations are reversed to get back the original image.

Boson sampling [162], a variant of hyper-chaotic systems, generates chaotic sequences through two key steps: scrambling and replacing. First, the Arnold transform is applied, followed by global position scrambling to randomize pixel positions. The grayscale pixels are then encrypted using XOR encryption and a block-cyclic shift, enhancing security. Decryption reverses these scrambling and replacing steps. However, the model operates on 4×4 grayscale images on a programmable silicon photon processor entailing 5 modes and 2 photons.

Similarly, Verma et al. [163] use the quantum Arnold transform to scramble the image. Then, XOR operations with 3D-BNM chaotic map diffuse pixel values, and controlled qubit scrambling enhances randomness. C—NOT and CC—NOT gates secure encryption through selective scrambling, with simulations confirming resistance to attacks.

Rehman [164] developed an encryption algorithm for NEQR-based images using 2D quantum coding and a 1D sine-chaotic map. It overcomes the limitations of traditional 1D chaotic systems, providing enhanced security.

Prajapat et al. [165] employed chaotic map and gray coding techniques to encrypt healthcare images, generating encryption keys using the logistic sine map for enhanced security.

Furthermore, the original BRQI-based image is encrypted through bit-plane permutation, creating an intermediate cipher image. It is then XORed with a quantum key generated using the logistic chaotic map, followed by a row-column permutation to produce the final encrypted image [166].

As most approaches consider the encryption of grayscale images only, Ma et al. [167] extended encryption to multiple RGB images, building on the foundations in [160]. It uses Haar quantum wavelet and iterative quantum Fibonacci transforms to compress and scramble a hybrid image (made of 1 to 4 images), respectively. A quantum image diffusion scheme increases the randomness of integer sequences and creates a quantum key image. The key image is XORed with the scrambled image to generate the ciphertext quantum image.

3.4.2. Steganography

Quantum steganography exploits the properties of qubits for embedding confidential information in quantum carrier images (Fig. 13). The QBIP model [130] prioritizes the MSB for further processing due to its significant impact on the image's representation. Given that the LSB has a minimal effect on the original image, it finds practical application in steganography as alterations made within the LSB remain

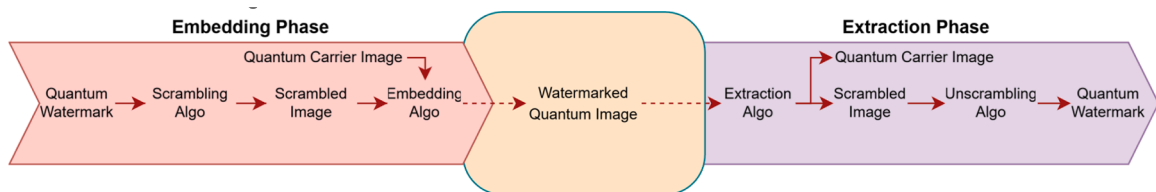


Fig. 14. In the embedding phase, a quantum watermark undergoes transformation through a quantum scrambling algorithm, producing a scrambled image. This scrambled image is then embedded into a quantum carrier image using a quantum embedding algorithm, resulting in a watermarked quantum image. In the extraction phase, a quantum extraction algorithm retrieves the scrambled watermark from the quantum carrier image. The unscrambling algorithm then reconstructs the original quantum watermark.

imperceptible to the human eye, making it an ideal choice for concealing information without arousing suspicion. Furthermore, the least significant qubit information-hiding algorithm [168], built on NEQR, demonstrated a higher PSNR for the standard Lena image.

Furthermore, Jiang *et al.* [169] proposed a Moiré pattern-based NEQR image steganography that uses quantum circuits to hide a binary image within a grayscale image. The process begins with selecting a stochastic Moiré grating as the cover image, which is then modified based on the secret image to create a deformed Moiré pattern. This pattern is further altered to produce the final steganography image. An enhanced version [24], employing two blind steganography algorithms using quantum circuits based on NEQR representation, has been proposed. The first algorithm employs standard LSB, where message bits directly replace the pixels' LSB. Though simple, the standard LSB system lacks robustness. The second algorithm, block LSB, improves robustness and undetectability by embedding a message bit into multiple pixels within an image block. Standard LSB can be viewed as a special case of block LSB, where each block contains only one pixel. Experimental results show that this approach maintains good invisibility and allows for an adjustable balance between capacity and robustness based on application needs [24].

Additionally, Wang *et al.* [168] proposed a least significant qubit (LSQb) information-hiding algorithm for NEQR images. It embeds a secret message qubit stream into the last qubit of the color information in a quantum cover image. They also introduced an LSQb frequency domain-based information hiding algorithm to enhance security. It applies a QFT to the cover image and then uses a two-qubit quantum comparator to compare the color encoding information qubits of the secret image with the last qubit of the QFT-transformed cover image. Based on the comparator's outputs, different unitary transformations are applied to embed the secret information into the QFTed cover image [168].

3.4.3. Watermarking

Quantum watermarking embeds signals (visible or invisible) into quantum multimedia (carrier) like images and videos that act as a signature for authenticating ownership and protecting image content and copyrights (Fig. 14) [170]. The embedding procedure relies on a confidential key, determining precise locations for insertion within the carrier. This same key, aligning with the below standards, enables subsequent extraction of the embedded watermark. Potdar *et al.* [171] delineate three fundamental imperatives that digital watermarking must fulfill to ensure security:

- 1) **Robustness:** ensuring the ability to extract the watermark even if it undergoes deliberate or inadvertent manipulation.
- 2) **Transparency:** depicting that the watermark can be added without distorting the image or requiring cropping.
- 3) **Data payload:** specifying the data indispensable for ensuring signal uniqueness and successful watermark extraction [171].

Nevertheless, the pioneering quantum image security protocol, WaQI [172], uses restricted geometric transformations for invisible watermarking, creating a secure, keyless, blind, and computationally efficient method to authenticate image ownership. It embeds a watermark into a cover image using a watermark embedding circuit, which is reversed to recover the original image for validation. Its grayscale version, gWaQI [173], employs a two-tier strategy, embedding a conspicuous watermark logo in a specific area and the same logo more obscurely across the rest of the image. The circuit embeds the visible and invisible watermarks in two stages.

However, the limitation of WaQI is the need for the watermark content to create the authentication circuit to validate ownership. Similarly, Zhang *et al.* [174] devised a watermarking technique to optimize the utilization of the carrier image's entirety for embedding the quantum watermark. An improvised WaQI model for MCRQI images is

based on generating double keys using quantum algorithms and embedding the watermark in the spatial and frequency domains [75]. The results showcased surpassing the base variant by getting higher PSNR values for testing images. Another extension of WaQI [66] utilizes the quantum Fourier transform that processes the watermark image through a quantum circuit to enhance its security, with the watermark embedded within Fourier coefficients. It ensures resistance against cropping or noise contamination. Experimental findings demonstrated notably high PSNR values.

Song *et al.* [67] introduced a dynamic quantum watermarking method using QWT for images based on FRQI to extract wavelet coefficients. A dynamic diagonal vector regulated embedding strength, offering greater adaptability than fixed parameters in classical methods. This vector, derived from the carrier and watermark images, accessible only to the authorized owner, served as a unique key for extraction. Optimized matrices ensured clear visual representation with high PSNR values in specific images. The scheme also reduced computational complexity compared to its classical variant. In addition, Yang *et al.* [175] improved Song *et al.*'s [67] quantum watermarking method, tackling challenges in adapting classical algorithms to quantum equivalents. Unlike classical methods, direct adjustment of coefficient grey levels for embedding watermark images was not feasible in the quantum domain. Yang *et al.* introduced a theoretical advancement employing a phase rotation operation, determining the degree of rotation for original carrier image pixels based on the embedding algorithm.

Furthermore, Soliman *et al.* [176] introduced an adaptive watermarking method using Weighted QPSO. Their approach outperformed traditional PSO and QPSO methods with a PSNR of 55 dB, tested against JPEG compression, cropping, and noise attacks. They suggested enhancing adaptability and performance by refining the algorithm's linear updating of weighted coefficients.

However, combining quantum encryption and watermarking, Zhou *et al.* [25] proposed a secure image transmission and identity authentication method. The watermark is scrambled using the QBM and embedded in the two middle-frequency sub-bands using the quantum Haar wavelet and discrete cosine transforms. Special preprocessing improves embedding capacity while keeping it invisible and robust. Initially, after representing the watermark and carrier images using the NEQR method, the 3D Henon hyperchaotic method generates chaotic sequences. The watermarked image is then scrambled multiple times using QBM, followed by applying QDCT. The carrier image is divided into four sub-band images using QHWT, with QDCT applied to the middle sub-bands. The scrambled watermarked image is then embedded into the carrier image, and the inverse QHWT and QDCT are used to extract the watermark.

Similarly, Zi *et al.* [177] introduce a quantum color image watermarking method that combines Quantum LOG edge detection with QDCT. Edge detection identifies the best channel for embedding the watermark, followed by decomposition with QHWT. QDCT is applied to embed the watermark in mid-frequency areas, enhancing resistance to JPEG compression.

Gorle *et al.* [178] convert the binary pixel values of the original and watermark images into qubits, with each 8-bit grayscale pixel represented by eight qubits. The watermark is embedded by combining the quantum circuits and applying CNOT gates to entangle the images, securing the watermark. Measurements are then performed to extract and process the watermark, reconstructing the watermarked image.

Li *et al.* [179] scrambled watermark image by altering both color values and positions, enhancing privacy. The scrambled watermark is then embedded into the carrier image's spatial domain using the LSB algorithm. Then, it is embedded into the frequency domain via quantum Fourier transform and LSB, boosting resistance to attacks.

Aly *et al.* [180] first converted the watermark and carrier color images to the quantum domain using the New Quantum Binary Image Detector [181], with each pixel represented by θ and ϕ parameters on Bloch spheres. These values act as keys for extraction. The quantum

Table 6
Summary of quantum machine/deep learning models.

Ref.	Models	Encoding/ Method	Dataset (s)	Classification Type		Image Type	Image Processing Method
				Binary	Multi		
Quantum Machine Learning							
[183]	QSVM	-	SemCity Toulouse	✓	-		QUBO
[184]		-	Handwritten digits (6 and 9)	✓	-		GRAPE, NMR
[185]			Handwritten characters	✓	-		
[186]	QKNN	Ground state	MNIST	✓	✓		Hamming distance
[187]		Superposition	Caltech-101 & Graz-101	-	✓		Color and text features
[188]		Amplitude	Handwritten digits	✓	✓		0–1 Matrix Transformation
[189]		Ground State	MNIST, FMNIST, & CIFAR-10.	✓	-	  	K-L Transform
[42]	QHL	Hamiltonian	Binary images for segmentation	-	-		PHLA+Gradient Descent for image segmentation
Quantum Deep Learning							
[190]	VGG16 + VQC	Angle	LFW-Gender	-	✓		ClassicalVGG16
[191]	ResNet18 + VQC		MRI Brain Tumor images	✓	-		Classical ResNet18
[192]	HNN		Damaged building images taken using a satellite.	✓	-		Classical CNN
[193]	QCNN		EuroSAT	-	✓		PQC
[194]	HDNN	Amplitude	UC Merced Land Use, AID, RESISC45	✓	-		quantum measurement-based features
[195]	QResNet18	Angle	BCDR dataset.	✓	-		Classical ResNet18
[196]	VQDNN	Hybrid	MNIST & UCI	✓	✓		Classical CNN
[197]	LLQNN	Angle	MNIST	✓	-		PQC with PCA
[198]	VQC		Malaria Cell Images	✓	-		Contourlet transform and contour areas
[199]	HQCNN		MNIST	✓	-		Classical CNN
[200]	QNN	Ground State	MNIST	-	✓		Quanvolutional filter
[201]	QDL		Steel images from Descaler	✓	-		Quantum Convolutional filter
[202]	HQCNN	Angle	MNIST, FMNIST, CIFAR-10	-	✓	  	HQconv, FQconv
[203]	VQC	Data re-uploading	MNIST	-	✓		Data re-uploading
[204]	QCNN	Amplitude	MNIST	-	✓		Classical CNN
[205]	VCNN		MNIST & FMNIST	-	✓		
[206]	QCNN		MNIST & FMNIST	-	✓		
[41]	QCNN	Hybrid Amplitude	MNIST & FMNIST	-	✓		
[129]	QMLSONN	Angle	Synthetic & Spanner images for denoising	-	-		MLSONN with back propagation

watermark is scrambled with a random reorder function, and the embedding process combines the scrambled watermark with the carrier image using a simple operation.

Unlike most methods that embed a single watermark, Xing *et al.* [182] propose a Multi-Modal quantum watermarking scheme with four modes for grayscale and color images. It incorporates the BBCE method for robustness against noise and geometric attacks alongside a brightness-based mechanism to minimize visual distortions. Three adaptive embedding strategies using controlled quantum gates are adapted to different modalities.

3.5. Quantum machine learning

Quantum machine learning leverages the computational power of quantum systems to accelerate machine learning tasks, offering significant potential in image processing. Models like QSVM, QKNN, and quantum deep learning optimize classification and pattern recognition in quantum images. A key advantage of quantum machine learning in image processing is its ability to efficiently process high-dimensional image data, overcoming the limitations of classical image processing. The latter encodes classical images into quantum states, while the former provides faster and more scalable solutions to tasks like image denoising, edge detection, and segmentation. A practical example of this integration is the work of Chin *et al.* [42], who developed circuits for quantum Hamiltonian learning using a parameterized Hamiltonian algorithm. This method converts binary image data into quantum states and applies PHLA, which accurately segments images by refining the loss function and updating circuit parameters through gradient descent. Such methods demonstrate how this integration can revolutionize tasks requiring complex image analysis at quantum speed. Models like QMLSONN [129] uses a neighborhood topology-based interconnection matrix along with quantum backpropagation to mitigate noise in quantum images. Further, the literature discussed in this section depicts the efficiency of quantum machine learning algorithms in accurately classifying different image data. However, a summary of the relevant literature is presented in Table 6.

3.5.1. QSVM

A novel QSVM, introduced by [183], tailored for remote sensing classification, explicitly focuses on urban area segmentation in France using the SemCity Toulouse dataset. Employing Quadratic Unconstrained Binary Optimization (QUBO) and Quantum Annealing (QA) principles, it aims to discern whether pixels correspond to building structures. Experimentation on d-wave and IBM quantum platforms showcased QSVM's superiority over classical SVM, achieving a notably higher F1 score. While the classical SVM attained an F1 score of 0.697, its quantum counterpart surpassed this performance with a score of 0.734. However, hardware limitations, such as qubit connectivity constraints, presented substantial challenges to maintaining coherence.

Furthermore, another variant of QSVM [184] distinguishes between the digits 6 and 9 using four qubits to outperform classical SVM. The digit images were transformed into a two-component feature space followed by normalization and linearization of the features. The architecture entailed Hadamard gates for superposition, matrix inversion, U_x gates, etc., to add feature vectors. Output predictions relied on ancilla qubit measurements using the ^{13}C spectrum. An upward peak (positive amplitude) signifies classification as "6", while a downward peak (negative amplitude) indicates "9". However, only eight images were tested, which is insignificant for robust evaluation. Upon relying on this architecture, Rana *et al.* [185] contrasted classical SVM and QSVM for handwritten character recognition. Initially, they classified the characters into "Baybayin" and "Latin" scripts and then classified different characters in each script. The results have shown that the QSVM outperforms classical SVM in terms of efficiency and accuracy improvement for both binary and multiclass classification.

3.5.2. QKNN

Leveraging a quantum circuit to compute Hamming distance, Ruan *et al.* [186] developed a QKNN algorithm to classify MNIST images. Unlike classical KNN, QKNN scales with feature size, improving efficiency. Training image features were encoded into 64-bit binary representations mapped to qubit states, while testing samples underwent a similar transformation. The quantum circuit used three registers for superposition, testing, and distance calculation. With $k = 20$, QKNN achieved nearly 98 % accuracy across different feature sets [186], outperforming classical KNN in classification performance. Dang *et al.* [187] extended this research by introducing a new distance metric for classifying RGB images from the Graz-01 and Caltech-101 datasets. They converted RGB images to the HSB (hue, saturation, brightness) to extract color features while integrating grayscale texture features. These features were flattened into 1-D arrays and encoded into quantum states for distance computation, where fidelity, measured via cosine similarity, served as the metric. Grover's search algorithm was employed to identify the minimum k values for classification, leading to an accuracy of 83.1 % on Graz-01 and 78 % on Caltech-101. A similar approach [188] enhances QKNN for handwritten digit recognition by leveraging quantum superposition to prepare quantum states for testing and training sets. Cosine similarity was computed using a quantum circuit, while distance information was encoded in quantum state amplitudes via the quantum amplitude estimation algorithm. However, the study did not provide specific details on the dataset or experimental setup. In contrast, Zhou *et al.* [189] proposed an advanced QKNN method employing the Karhunen-Loeve transform for feature extraction from RGB and grayscale images. They employed Hamming distance as the similarity metric and implemented a distance-weighted voting strategy for majority voting. Experiments on the MNIST, FMNIST, and CIFAR-10 datasets, conducted with varying k values revealed optimal performance at $k=30$. The proposed method achieved accuracies of 97 % on MNIST, 90 % on FMNIST, and 51 % on CIFAR-10, demonstrating notable improvements over classical KNN.

3.5.3. Quantum deep learning

VQCs and QCNNs use PQCs trained with classical optimizers to reduce errors, enhance learning, and unlock the potential of quantum computing in AI. VQCs share structural similarities with classical ANNs, employing quantum gates as trainable parameters to model complex functions and perform classification or regression tasks. On the other hand, QCNNs are inspired by classical CNNs, leveraging quantum gates for hierarchical feature extraction and dimensionality reduction [69]. These quantum models are particularly advantageous in handling high-dimensional data and demonstrating quantum advantage in learning tasks.

Identification of gender over facial images was conducted using classical to quantum transfer learning by [190], integrating a pre-trained VGG-16 with a VQC layer. The evaluation yielded a 93 % identification rate, although details of encoding and VQC architecture were limited. Relying on the methodology of [190], T *et al.* [191] used ResNet-18 for feature extraction and VQC for the classification of brain tumor images, resulting in a validation accuracy of 83.66 %, albeit without comparison to classical models. Furthermore, adding a superficial VQC layer to a Hybrid CNN was used to propose a hurricane damage detection technique [192]. However, the results did not overcome the classical CNN, as learning halted after 15 epochs. Employing three types of VQC structures within a hybrid model, [193] classified land types in remote sensing images. Upon evaluation, the amplitude circuit yielded the highest accuracy of 92 %, outperforming classical models. Nevertheless, multi-level classification yielded an accuracy of 97 %, showcasing superior efficiency compared to a ResNet-50 model with significantly fewer layers. Further, testing over the LULC semantic map of Beirut city revealed that the quantum classifier accurately distinguishes urban, vegetation, and water zones, performing comparably to models such as Wide-ResNet and JEM.

Merging quantum feature extraction and classical deep neural networks for image classification was conducted on an IBM quantum computer employing the amplitude encoding circuit [194]. The model was evaluated over three datasets (UC Merced Land-Use, AID, and NWPU-RESISC45), resulting in 95.89 %, 86.13 %, and 79.32 % accuracies, respectively. The quantum-classical framework outperformed state-of-the-art models while utilizing significantly fewer parameters. Azevedo *et al.* [195] enhanced classical transfer learning models by integrating a Dressed QuantumNet containing a VQC for breast cancer classification. It entails trainable parameters for fine-tuning, angle encoding, and multi-layered VQC for data processing. Realized on an IBM quantum computer, the approach outperformed other models in accuracy and AUC. Notably, the ResNet model combined with the Dressed QuantumNet outperformed not only its integration with DenseNet or ResNeXt but also classical models such as AlexNet, VGG19, DenseNet161, and ResNet18.

Wang *et al.* [196] proposed Variational Quantum Deep Neural Network (VQDNN) classifiers for binary and multiclass classification tasks, achieving 100 % accuracy in binary classification and 90 % accuracy for 10-class classification on MNIST and UCI datasets. The results demonstrated the potential of quantum models to improve performance with limited parameters and qubits. Addressing the Barren Plateaus problem in quantum networks, Skolik *et al.* [197] employed layerwise learning strategies, reducing generalization errors and showcasing the advantage of their approach in noiseless simulations compared to classical deep learning architectures.

Nevertheless, Hossain *et al.* [198] developed a quantum-enhanced system for malaria detection in blood cells, employing advanced image processing and quantum classification techniques. They collected infected and uninfected blood smear images, extracting only RBCs through preprocessing. A median filtering technique was applied to preserve edges, enabling enhanced feature extraction via the contourlet transform. Chromatin dots, key indicators for malaria detection, were identified using contour-based features. For dimensionality reduction, both PCA and Minimum Redundancy Maximum Relevance (mRMR) were utilized. The extracted features were then encoded using angle encoding and classified via a VQC. The model achieved a testing accuracy of 98.93 % with mRMR and 94.16 % with PCA, outperforming classical models and demonstrating the effectiveness of quantum-assisted medical image analysis.

Furthermore, Potempa and Porebski [199] analyzed classical and

quantum approaches to image classification, demonstrating the superiority of quantum models in performance, convergence speed, and stability.

A QuannvN, an extension of classical CNN with quantum convolutional filters, was proposed using a simple threshold-based encoding method [200]. This approach enhanced feature extraction by encoding images through multiple quantum filters. The model achieved 95 % accuracy on MNIST but relied on simplistic encoding and quantum state vectors, making it impractical for real quantum hardware [69]. Critiquing and addressing the limitations of the above work, Villalba-Diez *et al.* [201] investigated the integration of quantum filters in image quality classification, aiming to enhance efficiency in steel manufacturing. By incorporating variable weights in convolutional windows, they reduced training time at the cost of a slight decrease in F1 score. However, further research is needed to optimize quantum filter implementations. Jing *et al.* [202] advanced hybrid classical-quantum CNNs for RGB image classification, employing hierarchical and flat quantum convolutional layers with two ansatzes. Their model outperformed classical CNNs on MNIST, FMNIST, and CIFAR-10 datasets. Despite its effectiveness, the approach required substantial computational resources for simulating quantum circuits. Further, to address the complexity of multi-qubit circuits, Easom-Mccaldin *et al.* [203] proposed a single-qubit QCNN leveraging single-qubit encoding. The model achieved competitive accuracies on the MNIST dataset by employing data reuploading with only a single-qubit encoding. However, the approach did not utilize entanglement or multi-qubit superposition, limiting its potential quantum advantage. Additionally, once an optimal level of embedding is achieved, increasing circuit depth may introduce complexity with less improvement.

Furthermore, a novel approach using Approximation Quantum State Preparation (AQSP) encoding was proposed for QCNNs [204]. Applied to MNIST images, this method mitigates the time and depth constraints of conventional amplitude encoding, achieving an impressive 96.65 % accuracy. While effective, the isolation of AQSP from QCNN processes and the approximation nature of AQSP might limit accuracy. Likewise, Huang *et al.* [205] proposed a Variational Convolutional Neural Network (VCNN) to replace classical layers in CNNs, achieving superior performance in binary, three-class, and four-class classification tasks on MNIST and FMNIST datasets. Utilizing angle encoding and inspired by MERA quantum circuits, VCNN outperformed other quantum models, showcasing its potential. The model achieved an AUC of 99.71 % for

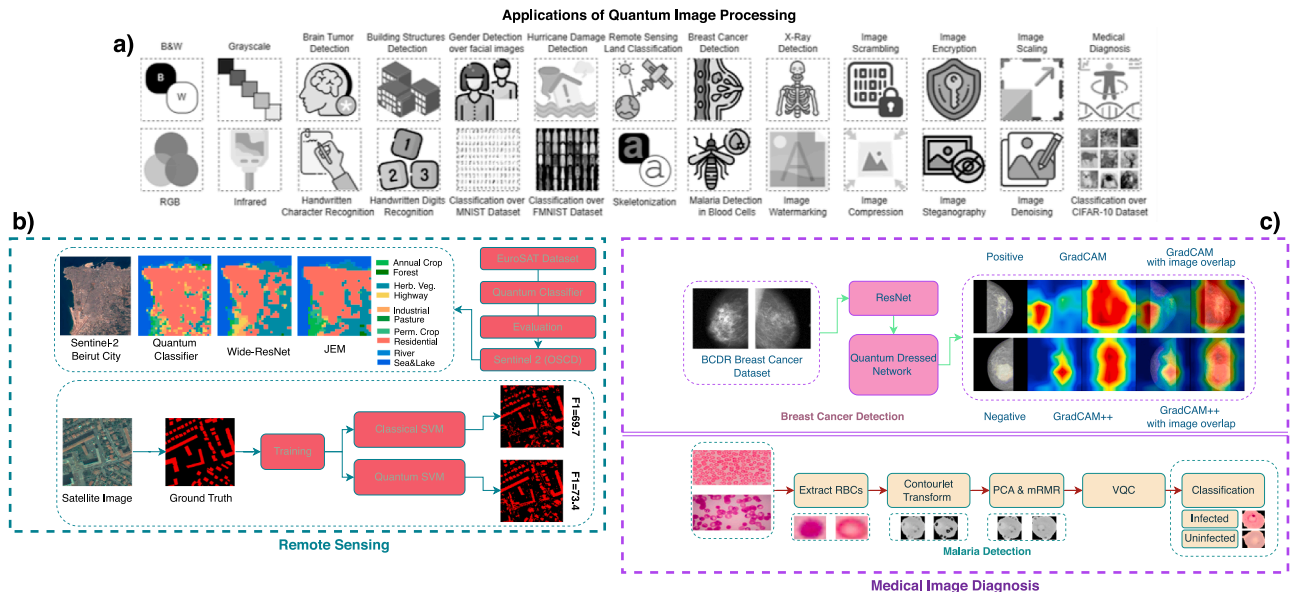


Fig. 15. (a) Key applications of quantum image processing, (b) Quantum remote sensing achieving superior classification accuracy over classical models, (c) Quantum machine learning for precise breast cancer diagnosis and malaria detection in blood cells.

binary (MNIST 0 and 1) and mean accuracies of 94.36 % and 92.49 % for multi-class classification (MNIST 0,1,3 and 0,1,36). For FMNIST, it attained 96.50 % accuracy for three classes (t-shirt, trouser, sandal) and 94.59 % for the four-class task, demonstrating its effectiveness. Furthermore, Hur *et al.* [206] explored various image reduction and encoding techniques for QCNNs, including PCA, autoencoders, amplitude, angle, and hybrid encoding methods. Their QCNN architecture, employing different ansatzes for convolution layers, achieved remarkable accuracies of 99 % on MNIST and 94 % on FMNIST datasets with minimal parameters. It highlights the effectiveness of QCNNs even with a small parameter space, although the absence of complex classification tasks could affect the reported accuracy levels.

Furthermore, Gong *et al.* [41] proposed QCNN incorporating a hybrid amplitude encoding method, combining angle and amplitude encoding for flexible and stable data encoding. Experiments on MNIST and FMNIST datasets show that QCNN achieves faster convergence (50 iterations vs. 200 for classical CNNs) and higher accuracy. For example, with PCA's reduction of features to 8, the proposed QCNN outperforms the classical CNNs in terms of accuracy.

3.6. Applications of quantum image processing

Quantum image processing transforms the handling and analysis of visual data, providing significant advantages over classical methods. There are various applications of quantum image processing, as can be seen in Fig. 15; however, the key ones include:

- 1) quantum image representation, which transforms classical images into quantum setting, along with quantum image encoding that offers compact representation of high dimensional image data for quantum machine learning [8,11,16–18,22,29,64,72–97,207,107];
- 2) quantum image compression, utilizing entanglement and superposition for more effective data reduction [8,16–18,22,64,72–97,207];
- 3) quantum filtering techniques that enhance image quality by applying advanced filters for noise reduction [19,208,111–120];
- 4) quantum edge detection and segmentation, crucial for fields like medical imaging, OCR detection, recognition, etc. [121–126,128–130];
- 5) quantum image transformations like morphological and geometric processing for enhancing quantum encryption algorithms by altering pixel values and positions [81,100–103,132–136];
- 6) quantum security like encryption, watermarking, and cryptography for data protection [20,22–24,30,40,56,57,65,119,140–155,157–172,209,210];
- 7) quantum machine and deep learning for remote sensing, satellite imaging, medical diagnostics classification, etc. [11,29,173,174,176–196,211].

Quantum image processing enhances data handling in engineering, medicine, and other applications, such as medical imaging, etc. It offers potential for improvements in computer vision, detection, and manufacturing. Quantum computing can also aid in big data, logistics, and cloud computing. In biopharmaceuticals, it can speed up drug discovery, while in AI, quantum machine learning improves tasks like pattern recognition and decision-making. Quantum optical imaging exploits light's quantum properties for high-resolution images, beneficial in areas like microscopy and astronomy. Despite these advancements, challenges such as hardware limitations and the need for more efficient algorithms persist, highlighting the importance of ongoing research to unlock the full potential of quantum image processing.

4. DISCUSSION

This section presents quantum experimental research, open areas

and research gaps in quantum image processing algorithms, highlighting areas that require further exploration.

4.1. Quantum experimental research

Testing quantum image processing models in real-time depends on the number of qubits used. Currently, quantum image retrieval is constrained to 2×2 -pixel images [212], as larger images require exponentially more quantum resources. In small-scale quantum systems, two-qubit pure entangled states exhibit genuine quantum correlations (e.g., Bell states), while three-qubit GHZ and W class states enable more complex correlations. Several experimental approaches have been proposed for generating GHZ states, which can be important for quantum image processing. Notably, Bouwmeester *et al.* conducted the first experimental demonstration of three-photon GHZ entanglement [213]. Zou and Mathis [214] suggested implementation of three-level states using Rydberg atoms in microwave cavity QED, while Leibfried *et al.* [215] demonstrated implementation of 6-qubit 'Schrödinger cat' states using beryllium ions. Similarly, Wei *et al.* [216] explored superconducting qubits, Walther *et al.* [217] used photon polarization, and Fiorentino and Wong [218] proposed hyper-entangled photon states. Further, experimental observation of a three-photon polarization-entangled W-state is conducted by Eibl *et al.* [219], while the realization of a perfect three-qubit W-state in a superconducting qubit system is experimentally observed by Swain *et al.* [220]. The optimal platform for quantum image processing will depend on future advancements in N-particle entangled state generation. As the number of qubits increases, quantum systems become increasingly difficult to control and more susceptible to decoherence, noise, and hardware limitations. Although quantum computation is inherently probabilistic, scaling up intensifies the challenges of maintaining coherence and achieving reliable outcomes. Key challenges include implementing robust quantum error correction to mitigate noise, developing high-speed quantum communication networks for efficient image transmission, and designing energy-efficient quantum architectures for large-scale image manipulation. To overcome these challenges, advancements in qubit scalability and quality, fault-tolerant quantum hardware, and hybrid quantum-classical approaches are essential for scalable quantum image processing. Innovations are needed in quantum memory, high-resolution detectors, and real-time processing capabilities to drive advancements in quantum-enhanced imaging techniques, enabling its full potential in image and video processing.

4.2. Open areas

- 1) With the advent of quantum computing, intractable classical algorithms are expected to become solvable. While notable progress has been made, numerous applications remain underexplored, presenting research opportunities, including advanced areas such as image reconstruction and enhancement, computer vision, high-resolution imaging, and semantic analysis to understand and interpret image content more deeply.
- 2) In reviewing quantum image processing algorithms and protocols, it's evident that they predominantly rely on PSNR to evaluate image similarity. However, the unique quantum nature of quantum image processing challenges the suitability of PSNR for measuring fidelity between quantum images. Various quantum similarity matching models were introduced to fill this gap [221–224]. While promising, further refinement and experimental validation are necessary to establish their reliability as image quality metrics for quantum computers.
- 3) Despite extensive advancements in quantum image processing algorithms, the physical preparation and retrieval of quantum images remain underexplored, particularly in experimental settings. Given that quantum images are highly susceptible to decoherence and gate errors, integrating quantum error correction methods like repetition

codes, Shor, or surface codes could be used to mitigate errors in practical deployment.

- 4) While quantum image processing has garnered significant attention, research on text, audio, and movies representing diverse signal types remains scant, indicating a need for exploration and innovation. One promising direction is embedding audio signals into quantum images/videos, which can enable advanced applications in quantum steganography. By encoding audio amplitudes into quantum image states, secure and multimodal quantum information hiding techniques could be developed for enhanced security.

4.3. Research gaps

Numerous areas within the quantum domain still require in-depth exploration to advance image processing applications.

- 1) Various quantum image representations struggle with RGB color images, operate only on restricted image dimensions (like square formats ($2^d \times 2^d$)), can handle only a single image simultaneously, and cannot perform complex color operations. The quantum circuits used for continuous and phase representations, heavy in-depth, could not be executed for images with larger dimensions (like 32×32 , 64×64 , or higher). Quantum computers face errors from decoherence and noise, and current image representation techniques often lack effective error correction. There are limited real-world applications and practical demonstrations of quantum image representation techniques. Most research is theoretical or limited to small-scale experiments, which restricts understanding of these methods' practical benefits and limitations. It is computationally economical to simulate the encoding of high-resolution images, but there is still a gap for optimized algorithms that minimize resource usage while maintaining high fidelity. Quantum computation's advantage lies in efficiently mapping classical data into quantum superposition states. QRAM [225,226], which uses a "bucket brigade" structure to distribute N d -dimensional vectors across N^d leaf nodes, achieves superposition in $O(\log(Nd))$ time. However, its $O(Nd)$ resource demand scales exponentially, questioning its practicality in real experimental settings.
- 2) Nevertheless, quantum movie representation lacks exploration with color images and error correction, challenging data fidelity. For quantum audio representation, a gap highlighted in positive amplitude treatment urges exploration of error mitigation. Quantum movie and audio representations remain almost unexplored, presenting a valuable opportunity for breakthrough research that could transform how we process and experience digital media. Assessing its real-world efficacy in music and speech applications and suitability for audio tasks can enhance quality and perception, refining quantum audio processing and advancing audio fields.
- 3) In quantum scaling operations, bilinear interpolation algorithms remain relatively rare, while those employing bicubic interpolation are yet to be proposed. Certain quantum variants of classical image transformation algorithms generally fail to surpass the efficiency of their classical counterparts. Certain existing methods extend nearest-neighbor interpolation to quantum computing but produce lower-quality results than classical methods like bilinear and bicubic interpolation. Quantum algorithms for image translation are limited to entire translations and cyclic translations for $2^d \times 2^d$ grayscale images, with increased complexity for different directional translations. Initial implementations of quantum geometric transformations show promise but remain computationally demanding. Simulations of quantum geometric transformations face challenges due to memory and space limitations when run on classical computers, affecting the accuracy and efficiency of the results. The current quantum image transformation algorithms are complex, limiting their practical applications. Additionally, many other types of traditional image transformations remain unexplored in quantum settings. However, designing their quantum variants can enhance quantum image processing capabilities, enabling their effective application in quantum image security.
- 4) Current quantum denoising methods often lack versatility in handling different types of noise, highlighting the need for more adaptable denoising strategies. Various noise types remain unexplored, such as amplitude damping, phase damping, etc. Enhanced optimization techniques are required to improve the performance and efficiency of these algorithms. Additionally, there is a need for adaptive filtering strategies and robustness to environmental variations. Nevertheless, topological quantum computing [227] offers a promising path to eliminating quantum noise. Remarkably, certain unlearnable functions in noisy classical settings can become learnable in noisy quantum environments, like DNF [228], with research now extending to broader linear functions [229]. Adapting these approaches to quantum images can play a crucial role in error mitigation.
- 5) Most encryption algorithms focus on grayscale images, while color, double-quantum, and multiple images are still relatively rare. While the quantum encryption theory is well-developed, implementing it in real-world scenarios is still challenging. Building reliable quantum communication networks and QKD systems requires overcoming various technical hurdles, such as maintaining the coherence of quantum states over long distances and integrating quantum components with existing classical communication infrastructure. QKD protocols transmit photons through optical fiber or free space. But, the range over which quantum signals can be reliably transmitted is limited due to signal loss, noise, and interference. QKD allows for the generation of secure cryptographic keys between two parties by exploiting the properties of quantum mechanics. However, efficiently distributing these keys to multiple parties in a network while maintaining security is non-trivial. Key management, including key generation, distribution, and storage, remains a significant challenge in practical quantum communication systems. Building and operating quantum communication infrastructure can be expensive, particularly on scale. To address these challenges, developing hybrid quantum-classical encryption schemes, improving quantum error correction, and advancing quantum repeaters for extended transmission distances are crucial. Additionally, optimizing resource-efficient QKD protocols can enhance security, making quantum encryption more practical and cost-effective for widespread adoption.
- 6) Enhancing the robustness of quantum watermarking against various types of noise (e.g., quantum bit-flip, phase-flip, depolarizing noise) remains a significant challenge. There are limited frequency-domain watermarking algorithms specifically designed for quantum systems. Designing quantum watermarking protocols that are secure against a broad spectrum of attacks, including quantum-specific threats such as quantum forgery, collusion attacks, and signal processing attacks, remains a significant challenge. Balancing the imperceptibility of the watermark with the fidelity of the quantum image data is complex. To address these challenges, integrating quantum error correction techniques with quantum watermarking can enhance noise resistance during transmission. Developing adaptive quantum watermarking methods based on dynamic encoding strategies can improve robustness against attacks. Further, integrating quantum encryption in quantum watermarking can enhance security of quantum watermarked images.
- 7) Practical implementation of quantum steganography systems involves intricate control of quantum states and precise execution of quantum operations, which is currently challenging. QKD, a primary application of quantum cryptography, is limited in range due to factors like photon loss in optical fibers, etc. Additionally, QKD systems often have low key generation rates, limiting their efficiency for high-speed data transmission. Quantum steganography systems

are theoretically secure but can still be vulnerable to practical attacks, especially those targeting physical implementations or auxiliary information. Quantum steganography requires high fidelity in quantum state manipulation, which is challenging due to errors, decoherence, and noise in quantum systems. Current quantum steganography techniques' embedding capacity and data rate limit their practical applications. Furthermore, the embedding of diverse signal types in quantum images remains largely unexplored, presenting a significant research opportunity for advancing quantum multimodal steganography.

- 8) The key challenges in the quantum machine and deep learning include efficient data encoding, scalability of quantum circuits, and demonstrating practical quantum speedup over classical methods. Other hurdles involve noise, error correction, and developing robust optimization techniques for training quantum models. Additionally, more sophisticated quantum deep learning architectures, hybrid classical-quantum models, and domain-specific applications are needed to advance the field effectively. Further exploration is critical to revolutionize the diagnosis of diseases like breast cancer, prostate cancer, brain tumors, etc. in quantum scenario. To address these challenges, optimizing quantum data encoding, implementing noise-adaptive training, and enhancing quantum error correction can improve model reliability.

5. Conclusion

In this study, we have systematically analyzed the recent state-of-the-art advancements in quantum image processing, including quantum image representations, advanced algorithms, transformative techniques, and robust security measures. While significant progress has been made, several critical research gaps remain, posing challenges to the practical realization of quantum image technologies. Overcoming barriers such as implementation complexity, interoperability, standardization, and scalable deployment is crucial for widespread adoption. Additionally, the robustness of quantum watermarking against noise, the security of quantum steganography systems, the balance between imperceptibility and fidelity, and the efficient use of quantum resources require further exploration. Further, the complexity of quantum key management, the vulnerability to attacks, and the limitations of current quantum cryptographic protocols underline the need for more secure and resilient approaches. Integrating quantum machine learning with image processing has shown promising results in classification and recognition tasks, yet more exploration is needed for more critical domains.

Quantum image processing can be extended to text, audio, and video while integrating with quantum machine learning and neural networks for tasks like NLP, cancer detection, genomic analysis, drug discovery, cybersecurity, and computer vision. Practical applications in quantum sensors, robotics, and nanoscale materials should be prioritized. Advancing scalable quantum algorithms for high-resolution images, optimizing resource usage, and enhancing error correction are crucial for real-world adoption. Interdisciplinary collaboration across quantum information science, cryptography, and engineering is essential for overcoming these challenges, driving quantum-driven breakthroughs in digital security, secure communication, and advanced quantum image processing.

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Data availability

The data availability is not applicable to this manuscript.

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